

Epithelial tissue in fish gills forms the half of the body surface area. The current study illustrates that gills are affected due to accumulation of silver nanoparticles causes epithelial lifting, hyperplasia and hypertrophy of epithelial cells. *Cyprinus carpio* reported a decrease in the blood glucose level when exposed silver nanoparticles which might be due to induction of hypoglycemia in the fish or due to the stress of toxicant. Significant decrease of glycogen may be due to the absence of metabolite named plasmatic pyruvate in liver. Current study reveals histological damage to gill surfaces by silver nanoparticles is attributed to high accumulations in gills, irritation due to elevated mucous secretion, increased ventilation. Biotransformation of toxicant and alteration of hepatic blood flow and potentiation of hepatotoxicity causes hepatic cell damage in liver or it might be due to distinctive lesions and the severity of the lesions seems to be related to the dose administered potency of silver nanoparticles. In the present study treatment of silver nanoparticles in kidney of fish *Cyprinus carpio* shows glomerular expansions, swell of tubules, dilation of capillaries.



Dr. Y. Thangam is working as an Assistant Professor at J.K. K. N. Recently she has received Sevai Chemmal award for serving the Covid affected persons.

## Silver Nano Particles in Freshwater Fish *Cyprinus Carpio*

Nanoparticles are small sized particles. In the aquatic environment nanoparticles depends on different types of factors



Y.Thangam Baskar  
S. Umamaheswari Sivaprakasam  
S. Kalaimani Subramani

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# ACCUMULATION OF HEAVY METALS IN RIVER BHAVANI

## INTRODUCTION

Nanoparticles are small sized particles. In the aquatic environment nanoparticles depends on different types of factor such as organic matter, pH, composition and ionic strength. These small sized particles leads to a high volume surface ratio metals and other compounds which in turn leads to a lot of binding sites (Moore 2006). Accumulation of nanoparticles enters through the membrane and accumulates inside the cell and causes damage to the cell (Choi *et al.*, 2008). Silver nanoparticles have been reported to have an effect on accumulation, DNA and development, reproducibility. In this study fresh water fish *Cyprinus carpio* showed effects on survival, embryonic growth and pigmentation at high concentration of AgNP which affects the swim bladder and larval swimming at low concentration of AgNP. Due to the exposure of silver nanoparticles neuro behavioral changes occur and this is the end points and those changes are different from Ag<sup>+</sup> (Powers *et al.*, 2011). AgNP act as catalyst and produce reactive oxygen species, ROS (Choi *et al.*, 2008, Moore 2006). The aim of the study is to determine the accumulation of heavy metals in different organs in fish *Cyprinus carpio* in river **Bhavani**. The metals Ag, AgNPs, Cd, Cr, Cu, Fe, Pb, Zn, Mg, in fresh water fish were evaluated.

The major river Bhavani is getting polluted by the chemicals, dyes, pesticides, fertilizers, drugs, heavy metals. The heavy metals like silver, silver nanoparticles, cadmium, chromium, copper, iron, lead, zinc, and magnesium were observed in this river. Many varieties of fishes were found in river bhavani. Fishes like *Cyprinus carpio*, *Cirrhinus mrigala*, *Catla catla*, *Labeo rohita*, *Tilapia mossambica* were found in this river. The above said metals accumulate in the fishes. Fish tissues are sensitive indicators of aquatic pollution (Mela *et al.*, 2007). The exposure of fish to these contaminants is likely to induce a number of lesions in different organs like gills (Cengiz and Unlu, 2006), liver (Korkman *et al.*, 2009), kidney (Bucher and Hofer, 1993), intestine (Brunelli *et al.*, 2008), Muscle (Costa *et al.*, 2009). These metals are absorbed by the gills and digestive tract and accumulate in the internal organs and tissues (Levesque *et al.*, 2003). Gills are the primary organ for respiration, ionic and osmoregulation, acid base balance. The gills in fish are a valuable model for assessing the effects of toxicants on cells and tissues (Mallatt *et al.*, 1995). Liver and muscle

plays an important role in vital functions, is the basic metabolism and it is the major organ of accumulation, biotransformation and excretion of contaminants in fish (Wu *et al.*, 2008). Kidney and intestine is the predominant organ involved in the regulation of intra and extra cellular volume and control of electrolytes and acid-base balance and major site of foundation of hormones that influence systemic metabolic functions. The metals like silver, silver nanoparticles, cadmium, chromium, copper, iron, lead, zinc, and magnesium are of particular interest because they are toxic to aquatic organisms especially to fish *Cyprinus carpio* and are persistent in the environment in excess and they accumulate in different organs and many changes were observed and were discussed.

## MATERIALS AND METHODS

**SAMPLE COLLECTION:** Fish samples from both sites were caught using gill nets and analysis were carried out according to APHA (2005). The length and weights of each fish were measured. The fishes were washed using deionized water and placed in separate polyethylene bags with ice. The samples were kept frozen at  $-20^{\circ}\text{C}$  until ready for analyses. For water sampling, polyethylene bottles were acid-washed with 10% concentrated nitric acid  $\text{HNO}_3$  (v/v) and rinsed thoroughly with distilled deionized water. All glassware and equipment used were acid washed. The bottles were rinsed three times and immersed about 10 cm below the water surface before sampling. Three 500 ml of water samples were taken at each sampling point, and kept in ice while been transported to laboratory. The samples were filtered through  $0.45\ \mu\text{m}$  microspore membrane filter and acidified with concentrated  $\text{HNO}_3$  (65%) to a pH less than 2. The samples were kept at  $4^{\circ}\text{C}$  before treatment.

**SAMPLE PREPARATION FOR WATER:** About 9ml of concentrated  $\text{HNO}_3$  were added to filtered water sample and heated gently at  $70^{\circ}\text{C}$  until the solution become transparent. The solutions were allowed to cool and filtered using  $0.45\ \mu\text{m}$  micropore membrane filter. The solutions were then added up with ultra-pure of water to 100ml and analyzed for trace metal concentration.

**SAMPLE PREPARATION OF FISH:** Sample preparation of fish: All the glassware and plastics were soaked overnight in 10% (v/v)  $\text{HNO}_3$  and rinsed with distilled water and deionized water and dried before being used. With help of stainless steel, removed 5 gram of muscle tissue, gill, liver, intestine, kidney then it was kept in the mixture ( $\text{H}_2\text{O}_2 + \text{HNO}_3$ ) at

1:3 ratios for digested to a strong acid digestion and at 150. C for 20 minutes and allowed to cool at room temperature. Samples were processed in duplicate and then diluted to a total 50ml with ultra-pure water and filtered through 0.45  $\mu$ m microspore membrane filter paper for analyses.

## 1. REAGENTS

All reagents were of analytical reagent grade. Ultra-pure water was used for all dilutions. The element standard solutions from Perkin Elmer that were used for the calibrations were prepared by diluting stock solutions of mg/mL.

## 2. HEAVY METAL ANALYSIS

All samples were digested in CONC HNO<sub>3</sub> and H<sub>2</sub>O<sub>2</sub> in a beaker. The samples were diluted with ultra-pure water until 50 ml. After filtration, the prepared samples were determined for Cd, Cr, Cu, Fe, Pb, Zn, Mg, by using atomic absorption spectrophotometer (AAS) and Inductively Coupled Plasma (ICP-MS). Heavy metals included in this study are: Silver (Ag), Silver nanoparticles (AgNp), Cd (Cadmium), Cr (Chromium), Cu (Copper), Fe (Iron), Pb (Lead), Zn (Zinc), Mg (Magnesium), Element standard solution from Perkin Elmer was prepared by diluting stock solutions of 100mg/mL.

## RESULT

**Table 1.** represents the accumulation of metals, Ag, AgNps, Cd, Cr, Cu, Fe, Pb, Zn, Mg, in water and the lethal concentration of metal were determined as 1.0, 1.4, 1.5, 1.7, 1.0, 8.5, 5.7, 4.2, 1.0. Accumulation of different metals in fish *Cyprinus carpio* in gill, liver, intestine, kidney and muscle were evaluated in percentage and the changes were noted. The accumulation in liver and kidney were high in percent when compared to the gill, intestine and muscle. The liver and kidney plays an important role as liver converts glucose into glycogen, thus accumulation of nanoparticles is more in liver and the kidney receives the salts and balance the body, so the percent of accumulation of silver nanoparticles is high in fish.

River bhavani consists of variety of fishes. In these fishes *Cyprinus carpio* is taken for the study period. This is due to that the fishes were available throughout the year. So this is taken for research study and the accumulations of different metals in different organs were observed.

### Accumulation of Metals

1. Silver
2. Silver nanoparticles
3. Cadmium
4. Chromium
5. Copper
6. Iron
7. Lead
8. Zinc
9. Magnesium

**TABLE 1: ACCUMULATION OF METALS IN RIVER BHAVANI**

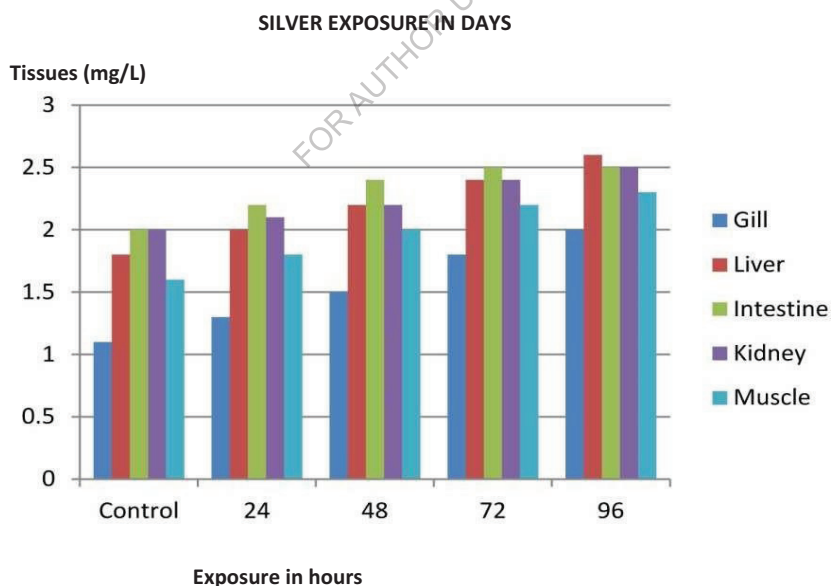
| S. No. | Metals | Accumulation of Metals |
|--------|--------|------------------------|
| 1      | Ag     | 0.75                   |
| 2      | AgNps  | 1.0                    |
| 3      | Cd     | 1.4                    |
| 4      | Cr     | 1.5                    |
| 5      | Cu     | 1.7                    |
| 6      | Fe     | 1.0                    |
| 7      | Pb     | 8.5                    |
| 8      | Zn     | 5.7                    |
| 9      | Mg     | 4.2                    |

**Bioaccumulation of metals and their concentration in organs to freshwater fish  
*Cyprinus carpio***

**TABLE 2 : Exposure of silver in fish *Cyprinus carpio* exposed to 24, 48, 72, 96 hours and accumulation observed in different organs like gill, liver, intestine, kidney and muscle.**

| Tissues   | Control | 24  | 48  | 72  | 96  | Percent Change |
|-----------|---------|-----|-----|-----|-----|----------------|
| Gill      | 1.1     | 1.3 | 1.5 | 1.8 | 2.0 | 55             |
| Liver     | 1.8     | 2.0 | 2.2 | 2.4 | 2.6 | 74             |
| Intestine | 2.0     | 2.2 | 2.4 | 2.5 | 2.5 | 76             |
| Kidney    | 2.0     | 2.1 | 2.2 | 2.4 | 2.5 | 72             |
| Muscle    | 1.6     | 1.8 | 2.0 | 2.2 | 2.3 | 67             |

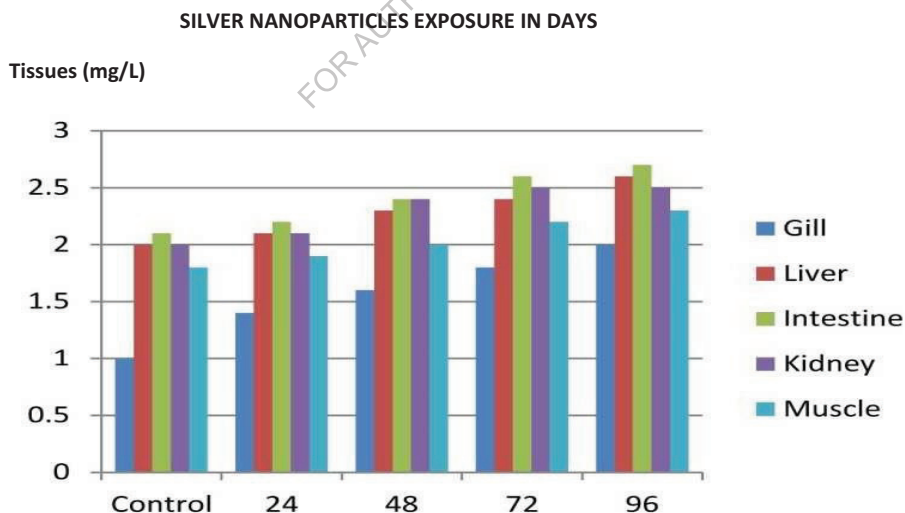
**Fig 1.**Represents the accumulation of *Cyprinus carpio* exposed to silver at 24, 48, 72, 96 hours concentration.



**TABLE 3 : Exposure of silver nanoparticles in fish *Cyprinus carpio* exposed to 24, 48, 72, 96 hours and accumulation observed in different organs like gill, liver, intestine, kidney and muscle.**

| Tissues   | Control | 24  | 48  | 72  | 96  | Percent Change |
|-----------|---------|-----|-----|-----|-----|----------------|
| Gill      | 1.0     | 1.4 | 1.6 | 1.8 | 2.0 | 59             |
| Liver     | 2.0     | 2.1 | 2.3 | 2.4 | 2.6 | 74             |
| Intestine | 2.1     | 2.2 | 2.4 | 2.6 | 2.7 | 78             |
| Kidney    | 2.0     | 2.1 | 2.4 | 2.5 | 2.5 | 75             |
| Muscle    | 1.8     | 1.9 | 2.0 | 2.2 | 2.3 | 66             |

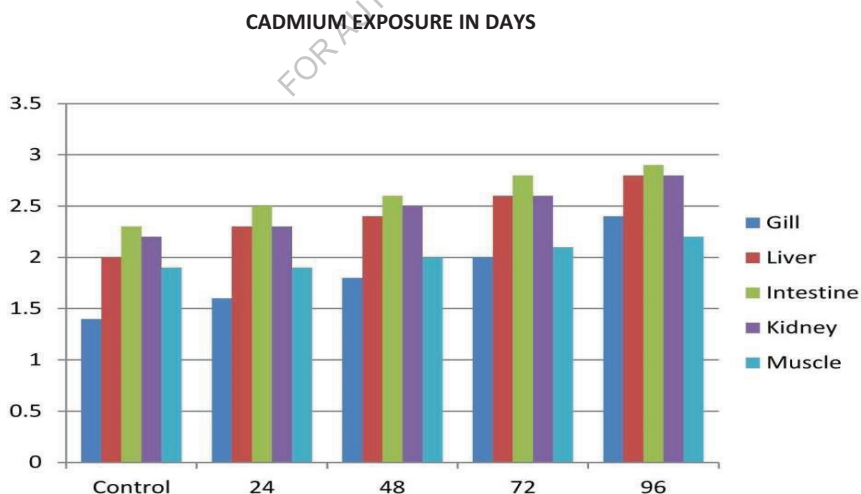
**Fig 2. Represents the accumulation of *Cyprinus carpio* exposed to silver nanoparticles at 24, 48, 72, 96 hours' concentration.**



**TABLE 4 : Exposure of cadmium in fish *Cyprinus carpio* exposed to 24, 48, 72, 96 hours and accumulation observed in different organs like gill, liver, intestine, kidney and muscle.**

| Tissues   | Control | 24  | 48  | 72  | 96  | Percent Change |
|-----------|---------|-----|-----|-----|-----|----------------|
| Gill      | 1.4     | 1.6 | 1.8 | 2.0 | 2.4 | 64             |
| Liver     | 2.0     | 2.3 | 2.4 | 2.6 | 2.8 | 81             |
| Intestine | 2.3     | 2.5 | 2.6 | 2.8 | 2.9 | 85             |
| Kidney    | 2.2     | 2.3 | 2.5 | 2.6 | 2.8 | 80             |
| Muscle    | 1.9     | 1.9 | 2.0 | 2.1 | 2.2 | 63             |

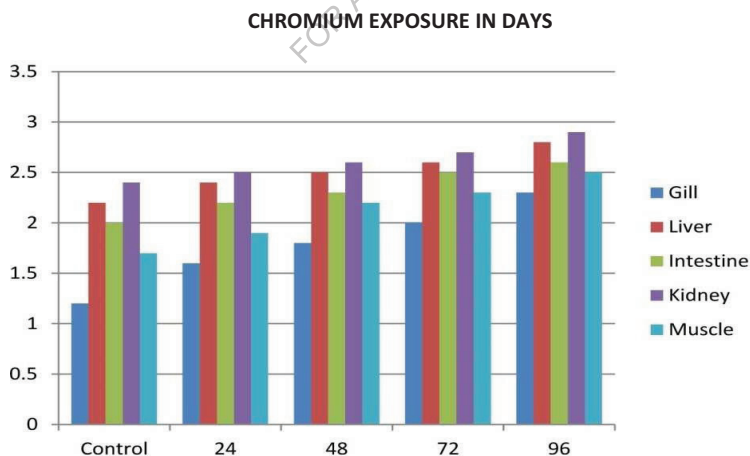
**Fig 3.**Represents the accumulation of *Cyprinus carpio* exposed to cadmium at 24, 48, 72, 96 hours concentration.



**TABLE 5 : Exposure of chromium in fish *Cyprinus carpio* exposed to 24, 48, 72, 96 hours and accumulation observed in different organs like gill, liver, intestine, kidney and muscle.**

| Tissues   | Control | 24  | 48  | 72  | 96  | Percent Change |
|-----------|---------|-----|-----|-----|-----|----------------|
| Gill      | 1.2     | 1.6 | 1.8 | 2.0 | 2.3 | 66             |
| Liver     | 2.2     | 2.4 | 2.5 | 2.6 | 2.8 | 81             |
| Intestine | 2.0     | 2.2 | 2.3 | 2.5 | 2.6 | 76             |
| Kidney    | 2.4     | 2.5 | 2.6 | 2.7 | 2.9 | 83             |
| Muscle    | 1.7     | 1.9 | 2.0 | 2.3 | 2.5 | 72             |

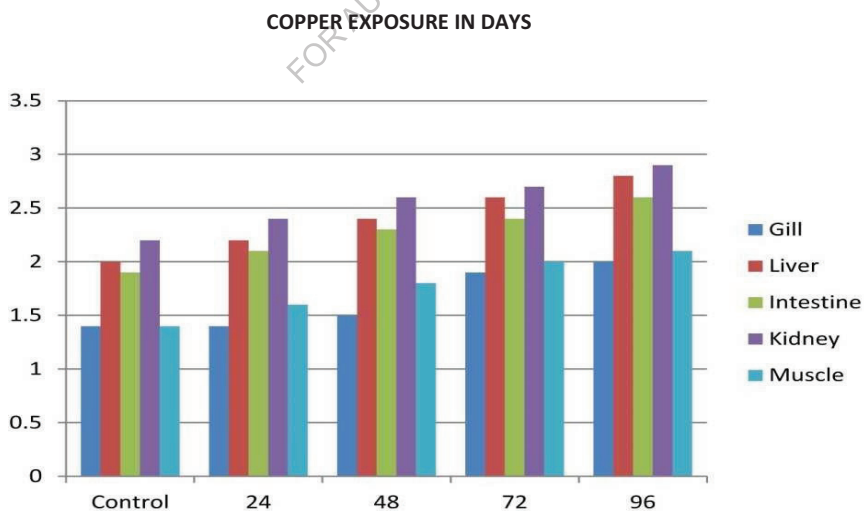
**Fig 4. Represents the accumulation of *Cyprinus carpio* exposed to chromium at 24, 48, 72, 96 hours concentration.**



**TABLE 6 : Exposure of copper in fish *Cyprinus carpio* exposed to 24, 48, 72, 96 hours and accumulation observed in different organs like gill, liver, intestine, kidney and muscle.**

| Tissues   | Control | 24  | 48  | 72  | 96  | Percent Change |
|-----------|---------|-----|-----|-----|-----|----------------|
| Gill      | 1.4     | 1.4 | 1.5 | 1.9 | 2.0 | 54             |
| Liver     | 2.0     | 2.2 | 2.4 | 2.6 | 2.8 | 80             |
| Intestine | 1.9     | 2.1 | 2.3 | 2.4 | 2.6 | 75             |
| Kidney    | 2.2     | 2.4 | 2.6 | 2.7 | 2.9 | 84             |
| Muscle    | 1.4     | 1.6 | 1.8 | 2.0 | 2.1 | 61             |

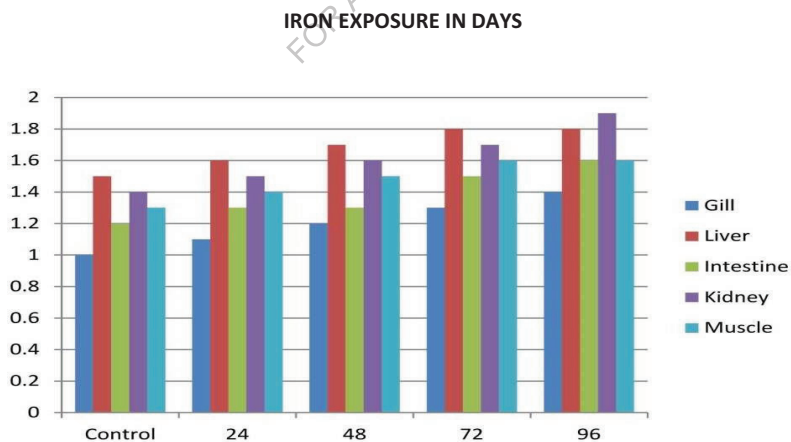
**Fig 5.Represents the accumulation of *Cyprinus carpio* exposed to copper at 24, 48, 72, 96 hours concentration.**



**TABLE 7 : Exposure of iron in fish *Cyprinus carpio* exposed to 24, 48, 72, 96 hours and accumulation observed in different organs like gill, liver, intestine, kidney and muscle.**

| Tissues   | Control | 24  | 48  | 72  | 96  | Percent Change |
|-----------|---------|-----|-----|-----|-----|----------------|
| Gill      | 1.0     | 1.1 | 1.2 | 1.3 | 1.4 | 40             |
| Liver     | 1.5     | 1.6 | 1.7 | 1.8 | 1.8 | 54             |
| Intestine | 1.2     | 1.3 | 1.3 | 1.5 | 1.6 | 45             |
| Kidney    | 1.4     | 1.5 | 1.6 | 1.7 | 1.9 | 53             |
| Muscle    | 1.3     | 1.4 | 1.5 | 1.6 | 1.6 | 48             |

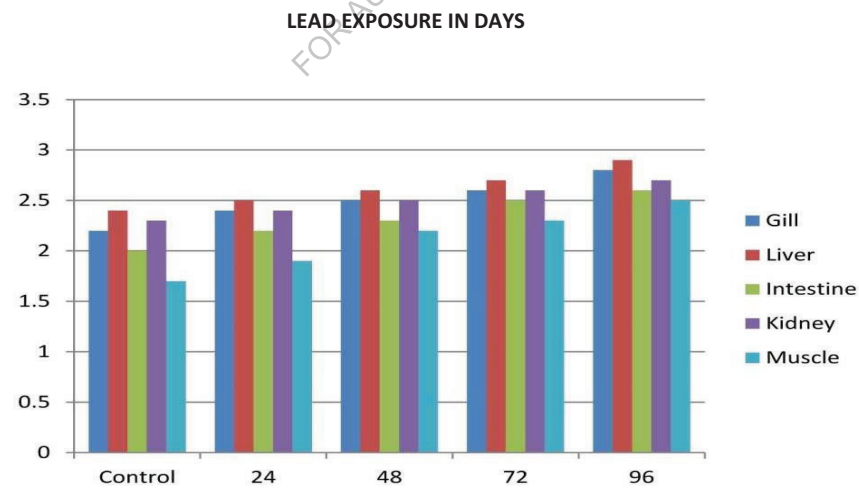
**Fig 6.Represents the accumulation of *Cyprinus carpio* exposed to iron at 24, 48, 72, 96 hours concentration.**



**TABLE 8 : Exposure of lead in fish *Cyprinus carpio* exposed to 24, 48, 72, 96 hours and accumulation observed in different organs like gill, liver, intestine, kidney and muscle.**

| Tissues   | Control | 24  | 48  | 72  | 96  | Percent Change |
|-----------|---------|-----|-----|-----|-----|----------------|
| Gill      | 2.2     | 2.4 | 2.5 | 2.6 | 2.8 | 83             |
| Liver     | 2.4     | 2.5 | 2.6 | 2.7 | 2.9 | 83             |
| Intestine | 2.0     | 2.2 | 2.3 | 2.5 | 2.6 | 76             |
| Kidney    | 2.3     | 2.4 | 2.5 | 2.6 | 2.7 | 79             |
| Muscle    | 1.7     | 1.9 | 2.2 | 2.3 | 2.5 | 72             |

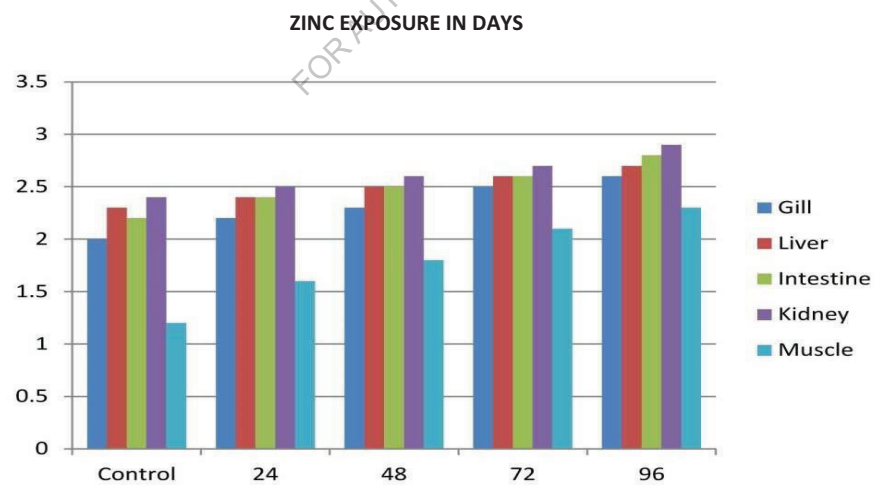
**Fig 7.**Represents the accumulation of *Cyprinus carpio* exposed to lead at 24, 48, 72, 96 hours concentration.



**TABLE 9 : Exposure of zinc in fish *Cyprinus carpio* exposed to 24, 48, 72, 96 hours and accumulation observed in different organs like gill, liver, intestine, kidney and muscle.**

| Tissues   | Control | 24  | 48  | 72  | 96  | Percent Change |
|-----------|---------|-----|-----|-----|-----|----------------|
| Gill      | 2.0     | 2.2 | 2.3 | 2.5 | 2.6 | 76             |
| Liver     | 2.3     | 2.4 | 2.5 | 2.6 | 2.7 | 79             |
| Intestine | 2.2     | 2.4 | 2.5 | 2.6 | 2.8 | 81             |
| Kidney    | 2.4     | 2.5 | 2.6 | 2.7 | 2.9 | 86             |
| Muscle    | 1.2     | 1.6 | 1.8 | 2.1 | 2.3 | 66             |

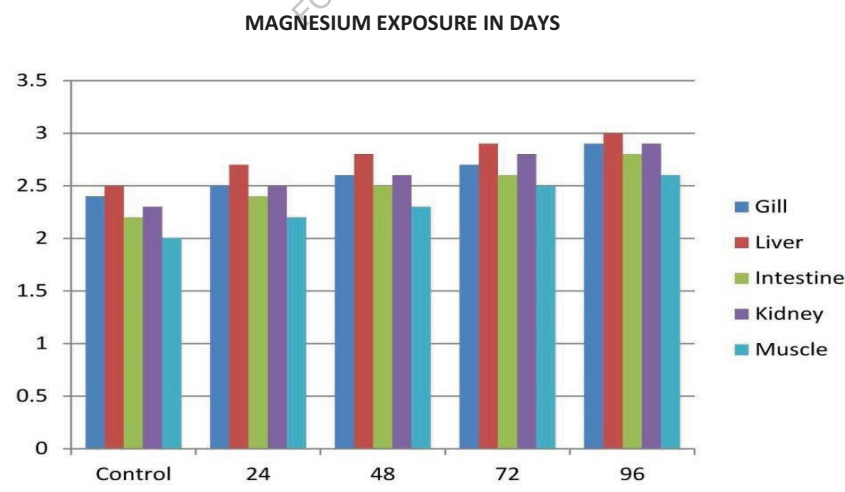
**Fig 8.**Represents the accumulation of *Cyprinus carpio* exposed to zinc at 24, 48, 72, 96 hours concentration.



**TABLE 10 : Exposure of magnesium in fish *Cyprinus carpio* exposed to 24, 48, 72, 96 hours and accumulation observed in different organs like gill, liver, intestine, kidney and muscle.**

| Tissues   | Control | 24  | 48  | 72  | 96  | Percent Change |
|-----------|---------|-----|-----|-----|-----|----------------|
| Gill      | 2.4     | 2.5 | 2.6 | 2.7 | 2.9 | 83             |
| Liver     | 2.5     | 2.7 | 2.8 | 2.9 | 3.0 | 89             |
| Intestine | 2.2     | 2.4 | 2.5 | 2.6 | 2.8 | 81             |
| Kidney    | 2.3     | 2.5 | 2.6 | 2.8 | 2.9 | 85             |
| Muscle    | 2.0     | 2.2 | 2.3 | 2.5 | 2.6 | 76             |

**Fig 9.Represents the accumulation of *Cyprinus carpio* exposed to magnesium at 24, 48, 72, 96 hours concentration.**



## DISCUSSION

On world wide scale metals and anthropogenic chemicals have resulted ecotoxicological effects. The heavy metals when released in higher concentration cause deleterious effect on organisms in aquatic ecosystem. In aquatic environment pollutants like heavy metals causes metabolic stress and originates in organisms that lead to mortality (Shaffi *et al.*, 2001). Fish is the most susceptible to elemental contaminants in aquatic environment than any other in aquatic habitants and it has been widely used to assess the effect of human import on aquatic ecosystem, water quality deterioration and changes might result in selective elimination of most sensitive life stage of vulnerable fish species as high levels of metals discharged in aquatic ecosystems (Bervoet *et al.*, 2005).

The heavy metal silver nanoparticles accumulate in fishes in different organs. The term bioaccumulation is defined as the chemicals taken up by an organism either directly from exposure to a contaminated medium or by consumption of food containing the chemical. Whereas metals are concerned, it can be defined as the net accumulation of a metal in a tissue of interest or a whole organism that results from exposure. By multiple routes of exposure (diet and solution) metals accumulate and this affects the bioavailability and geochemical methods. Accumulation of metals and metalloids is of particular value as it acts as an exposure indicator since metals are not metabolized. Bioaccumulation is often a good integrative indicator of the chemical exposures of organisms in polluted ecosystems. At some bioavailability all trace metals are toxic. Bioavailable toxic metal may come under selection for changes in one or more physiological processes, thus aquatic organisms are exposed to the rate of efflux of detoxification of accumulated metal into a relatively processing level. Half of the body surface areas comprise fish gills with an epithelial lifting in the environment. As results of this close association between the water and blood, the gills are strongly affected by environmental pollutants (Wood *et al.*, 1990). Many authors suggest that changes occur in different organs due to accumulation. Due to accumulation of these metals in gills alterations like epithelial lifting, hyperplasia and hypertrophy of epithelial cells, oxygen uptake is impaired (Fernandes and Mazon, 2003). Liver and intestine causes the metabolic problems. Bile is affected and digestion stagnation occurs in the liver and intestine. The accumulation of pollutants in bile indicates damage to the hepatic metabolism (Fanta *et al.*, 2003). The heavy metals accumulate in kidney and changes were noticed and this may be due to renal excretion of under toxified toxicant molecule resulting in severe pathological changes

in haemopoietic tissues resulting severe necrosis, cloudy smelling of renal tubules, disintegration of interstitial tissue, pycnotic nuclei (Tilak *et al.*, 2001).

In the present study fish gills comprise more than half of the body surface area. The gills are affected due to accumulation of silver nanoparticles and it might be due to epithelial lifting, hypertrophy of epithelial cells. Liver and intestine gets damaged and it may be due to metabolic problems. The heavy metals accumulate in kidney; changes were noticed and may be due to renal excretion of toxicant molecule resulting in severe necrosis and changes in haemopoietic tissues.

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# BIOCHEMICAL

## INTRODUCTION

Nanotoxicology is a branch of bionanoscience which deals with the study of nanomaterials and their impact on living organisms' threat to the environment and to human beings. The nanotoxic effects quantitative evaluation on severity and frequency in relation to exposure of the organisms is also considered. In various fields, metal nanoparticles have been used such as consumer products, industrial applications, health care technology and biochemical activities (Masciangioli and Zhang 2003; Nohynek *et al.*, 2007) and they enter the environment through aquatic and terrestrial sources. Benn and Westerhoff, 2008 stated that the most commonly used nanoparticles. AgNPs used in aquatic organisms are released into the aquatic environment through fishes. Ecotoxicological effects have increased due to the long term emission of nanoparticles on a world wide scale (Handy and Shaw, 2007).

Agrahari *et al.* (2007) reported that to identify toxicity organs as well as general health status of animals, biochemical parameters are helpful and also provide warning signal is stressed organism. The effects of stressors and toxic substances are determined by biochemical parameters. Aquatic bodies are traditional recipients in industrial waste containing heavy metals, salts, chemicals and pesticides which when released in higher concentrations cause deleterious effect on organisms. Persistent in some causes metabolic stress is caused by the presence of pollutants like salts, heavy metals in aquatic ecosystem, which some tank lead to mortality of organism. Aquatic Ecosystem is the final sink among a wide range of environmental pollutants which affects the non-target aquatic organisms (Osman *et al.*, 2010).

Silver nanoparticles are formed from eventual aggravation of silver cluster under various conditions. Silver is formed by the action of laser during ablation of metallic silver or silver salts. Silver to 100µm breaks up to nanoparticles called silver nanoparticles. By ionisation of Ag and silver nanoparticles  $\text{Ag}^{2+}$ ,  $\text{Ag}^{3+}$  is formed. To identify the members of a series of clusters, mass spectrometry plays a key role which have proven to be challenging due to their relative instability. In mass spectra, isotopic envelopes are developed because of the two isotopes of silver. The three combinations of silver isotopes yield three peaks in mass spectrum. In aerone channels silver cluster  $\text{Ag}_8$  can be stabilised. Erionite belongs to a

group of minerals called zeolites which is naturally occurring fibrous mineral with cavities of diameter 0.63nm and length 1.5nm. With diameter of 0.25 nm, the cavities of erionite are connected by small windows. Uptake and bio availability are affected by the form of silver which is dependent on biochemical and physiochemical condition of the environment, bioavailability of silver depends on speciation.

In reductive state, silver is in metallic state or in sulphide complexes which is insoluble in water. In oxidative state, silver binds with bromide, chlorides and iodides. Silver thiolates are found in polluted water. Salinity increase in water results in the formation of silver-chloro complexes and decreases the silver uptake (Ratte 1999). Analysis of blood chemistry by fish physiologists can provide important information about the function of organs in fishes specifically to detect and assess the metabolic disturbances in aquatic environment and aquatic ecosystems. Plasma and serum reflects the physiologic state of animals because they are the products of intermediate metabolism (Artacho *et al.*, 2007; Firat and Kargin, 2010). Fish physiologists make use of blood chemistry normal value indices for evaluation of fish stress responses, nutritional condition, reproductive state, tissue damage due to handling procedures and health status (Congleton and Wagner, 2006).

Aquatic organisms are the bio indicators in the monitoring of aquatic environment. Biochemical parameters detect the potential adverse effects in fishes and also events of pollutant damage (Almeida *et al.*, 2002). The metabolic basis for conventional fishery like growths are determined by biochemical methods viz, the detection of stress and mode of action are biochemical methods. The detection of toxic chemical effects at earliest stage is suggested by biochemical and physiological diagnosis method. The deleterious effect of environmental pollutants (Lopez *et al.*, 2009) and useful biomarkers of exposure to aquatic pollutants (Bainy, 1996; Ahmad *et al.*, 2000) is understood from the biochemical mechanisms involved in cellular detoxification.

In fish brain carbohydrate breaks down into glucose after a series of chemical reaction. Carbohydrate is the important source of energy for fish (Soengas and Aldegunde, 2002; Lopez-Olmeda *et al.*, 2009). Carbohydrate and lipids are the major stress indicators in teleosts (Radhakrishnaiah *et al.*, 1992; Yildiz and Benli, 2004) and energy precursors for fishes. Fishes are glucose intolerant and glucose last for several hours which leads to hyperglycemia, an abnormal high blood glucose level and is related to fish species (Moon, 2001). Blood glucose and hepatic glycogen are rapid, as they are considered as the stress

response parameters. The sensitive indicator of environmental stress in fish is by analysing quantitative blood glucose and glycogen level (Teles *et al.*, 2003). The report of increase in blood glucose level under stressful conditions is shown by many investigators. By handling forced thermal shock, plasma glucose is monitored (Pickering *et al.*, 1993; Ramesh *et al.*, 1994; Teles *et al.*, 2003; Pane *et al.*, 2005; Gagnon *et al.*, 2006).

Increase in blood glucose under stressful conditions of handling, silver in fresh water fish (Rozan, 1997) silver nanoparticles in *Zebra fish* and *Cyprinus carpio* (Yeo and Kang, 2008) glucose level increases chromium and zinc in *Salmo gairdner* (Strik *et al.*, 1975; Watson and Moniterio *et al.*, 2005) in *Pacu* and *Tambacu* due to nitrite exposure (Moraes *et al.*, 2006), Mercury in *Puntiusconchoni* (Gill and Pant, 1981), nickel in *Tilapia nilotica* (Ghazaly, 1992) arsenic in *Channa punctatus* (Hota, 1995). Crude oil in *Clupea pallas* (Kennedy and Farrell, 2005), cadmium chloride in *Catla catla* (Shobha *et al.*, 2007), mercury chloride in *Catla catla* (Martin Deva Prasath and Arivoli, 2008) and selenium in *Clarias gariepanus* (Abdel-Tawab *et al.*, 2007). The significant increase in glucose level in fishes is reported when exposed to acute concentration of heavy metals. Mercury (Gill and Pant, 1981) in *Puntius conchoni*; manganese (Ghazaly, 1992) in *Tilapia nilotica*, arsenic (Hota, 1995) in *Channa punctatus*, Zinc (Watson and McKeon, 1976) in *Salmogairdneri*. However significant decrease in plasma glucose level was reported by Bhattacharya *et al.* (1987) in *Channa punctatus*, Pane *et al.* (2005).

Decrease of glucose due to silver and silver nanoparticles exposure in *Atlantic cod* (Siikavuopio and Sae them 2006). Decrease in plasma glucose level was reported by ( Seok-Jeong, *et al.*, 2014) in silver nanoparticles in fresh water fish *Cyprinus carpio*, Chowdhury *et al.* (2004) in *Oncorhyncus mykiss* exposed to cadmium, Pane *et al.*(2005) in *Oncorhynchus mykiss* exposed to nickel, Gagnon *et al.*, (2006) in *Oncorhynchus mykiss* and *Cyprinus carpio* exposed to copper. Decrease in plasma glucose level was reported by Bhattacharya *et al.* (1987) in *Channa punctatus* exposed to industrial pollutants, Din rainbow trout. das *et al.* (2004a) in *Catla catla* exposed to nitrite and Velisek *et al.*(2007) due to deltamethrin . Sometimes no significant changes in plasma glucose may be observed, because under stress the fish is rapidly consuming energetic substrates. Similar degrees were recorded by Chowdhury *et al.* (2004) in *Oncorhynchus mykiss* exposed to nickel. The toxicity of silver and nanoparticles in plasma glucose and glycogen is the main aim to be assessed in present investigates. In this study exposure of silver nanoparticles to fish *Cyprinus carpio* results in decrease of glucose and glycogen content during 7<sup>th</sup> to 35<sup>th</sup> days studies. The main aim of

present investigation is to assess the toxicity of silver and silver nanoparticles in plasma glucose and glycogen of an Indian major carp *Cyprinus carpio* using parameters. We can understand the mode of action, stress and response as suitable biomarkers for silver and silver nanoparticles toxicity.

## **MATERIALS AND METHODS**

### **BIOCHEMICAL STUDIES**

Glucose and glycogen content in plasma activity of *Cyprinus carpio* were estimated.

#### **ESTIMATION OF PLASMA GLUCOSE**

O-Toluidine method is used for the estimation of plasma glucose was estimated by (Cooper and Mc Danial, 1970).

#### **PRINCIPLE**

Glucose reacts with O-Toluidine in the presence of acetic acid to form a green colour derivative which is measured at 630nm by using UV Spectrophotometer.

#### **REAGENT UTILIZED**

##### **Reagent**

1: O-Toluidine colour reagent Reagent

2: Glucose standard, 100 mg%

#### **PROCEDURE**

Tubes were taken and marked as Blank (B), Control (C), Test (T) and Standard (S). To each Test tube 5 ml of Reagent-1 (O-Toluidine colour reagent) was added. 0.1 ml of distilled water was added to the test tube marked 'B' (Blank). Similarly, 0.1ml of plasma from control and nitrite treated fish was added to test tube marked as 'C' (Control) and 'T' (Test) respectively. Then 0.1 ml of Reagent -2 (Glucose standard) was added to the test tube marked as S (Standard). The contents in the tubes were mixed well and heated in boiling water for 10 minutes. Then, the tubes were cooled under running tap water for 5 minutes and the optical density (O.D) of the test samples were measured at 630 nm within 30 minutes against Blank (B) using UV Spectrophotometer.

## CALCULATION

O.D. Of the test Plasma glucose in mg/100ml = ----- X 100 O.D. of the Standard IV.

## ESTIMATION OF PLASMA GLYCOGEN

Anthrone method is used for the estimation of glycogen (Samseifter *et al.*, 1949).

### PRINCIPLE

Sulphuric acid of the reagent causes dehydration of sugar to furfural derivatives which then presumably condenses with anthrone to form blue colour compound which is measured in UV Spectrophotometer at 620nm wavelength.

### REAGENT 1:

ANTHRONE REAGENT 0.2 g of anthrone and few crystals of thiourea were dissolved in 100ml of 95% sulphuric acid by using a standard flask.

### REAGENT 2:

30% POTASSIUM HYDROXIDE SOLUTION 30 g of potassium hydroxide was dissolved in distilled water up to 100ml in a standard flask.

### REAGENT 3:

STANDARD GLUCOSE 10 g of glucose was dissolved in distilled water up to 100 ml in a standard flask.

### PROCEDURE

The weighed tissues (liver and muscle) were dropped in separate test tubes marked as 'Control' (C) and 'Test' (T) tubes which containing 3ml of 30% potassium hydroxide solution (Reagent-2). The tissues were digested by heating the test tubes for 20 minutes. After cooling, the digest was transferred quantitatively to 100 ml standard volumetric flask and diluted up to the mark with distilled water. From this, 5ml of aliquots from each tissue digest were taken for estimation. 5ml of working standard glucose solution (Reagent 3) containing 100µg of glucose and blank (5ml of distilled water) were also prepared. Then 10 ml of the anthrone reagent (Reagent-1) was delivered from the fast flowing pipette and they were

mixed by swirling the test tube in a cold-water bath. Then they were heated for 10 minutes in boiling water and again cooled. Using UV Spectrophotometer, the optical densities (O.D) of all the tubes were measured against Blank (B) at 620 nm wavelengths.

## CALCULATION

$$\text{Glycogen in aliquots } 100(\mu\text{g/ml of digest}) = \frac{\text{O.D. Of the test } 100}{\text{O.D. of the Standard } 1.11} \times \text{X}$$

1.11 is the Morris conversion factor that is used for the conversion of glucose to glycogen.

## RESULT

Table 3 and Table 4 represent the data on changes in the biochemical profiles of fish *Cyprinus carpio* exposed to concentration of silver nanoparticles in glucose and glycogen level to sublethal toxicity. A significant decrease in the level of glucose and glycogen by silver nanoparticle is exhibited throughout study period. The significant decrease in glucose and glycogen level was directly proportional to the exposure period showing a percent decrease of -16.66, -12.17, -9.17, -7.54, -5.82 at the end of 7<sup>th</sup>, 14<sup>th</sup>, 21<sup>st</sup>, 28<sup>th</sup>, 35<sup>th</sup>, days for glucose level and for glycogen -4.18, -3.72, -2.70, -1.19, -1.24 at the end of 7<sup>th</sup>, 14<sup>th</sup>, 21<sup>st</sup>, 28<sup>th</sup>, 35<sup>th</sup>, days respectively. There were significant ( $P < 0.05$ ) variations among the treatments.

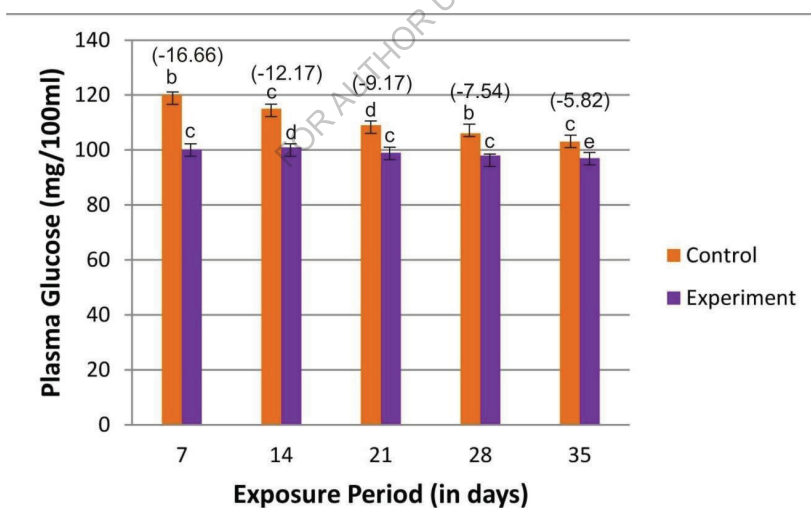
**TABLE 14 : Changes in the plasma glucose of *Cyprinus carpio* exposed to sublethal concentration of silver nanoparticles for 35 days.**

| S. No. | Exposure Period | Control         | Experiment   | Change % |
|--------|-----------------|-----------------|--------------|----------|
| 1      | 7               | 120 ± 0.099b    | 100 ± 1.375c | -16.66   |
| 2      | 14              | 115.00 ± 0.123c | 101 ± 1.097d | -12.17   |
| 3      | 21              | 109 ± 0.134d    | 99 ± 0.176c  | -9.17    |
| 4      | 28              | 106 ± 0.130b    | 98 ± 0.339c  | -7.54    |
| 5      | 35              | 103 ± 0.183c    | 97 ± 0.279e  | -5.82    |

|               |         |
|---------------|---------|
| Treatment (T) | 130.28* |
| Period (P)    | 87.19*  |
| TXP           | 110.10* |

Values are mean  $\pm$  S.E. of five individual observations. (+) Denotes percent increase over control. \* Significant at 5% level. Means in a column bearing same letter(s) are significantly different according to DMRT ( $P > 0.05$ )

**Fig. 12. Plasma glucose level of *Cyprinus carpio* exposed to sublethal concentration of silver nanoparticles for 35 days. Error bars indicate the standard error of the mean. Bars bearing same letter(s) are significantly different according to DMRT ( $P > 0.05$ ). The numerals in the parenthesis indicates percent change.**



**TABLE 15 : Changes in the plasma glycogen content of *Cyprinus carpio* exposed to sublethal concentration of silver nanoparticles for 35 days.**

| S. No. | Exposure Period | Control         | Experiment      | Change % |
|--------|-----------------|-----------------|-----------------|----------|
| 1      | C7              | 114.80 ± 0.046c | 110.00 ± 0.300b | -4.18    |
| 2      | 14              | 113.92 ± 0.270c | 109.68 ± 0.209c | -3.72    |
| 3      | 21              | 118 ± 0.051a    | 114.81 ± 0.278b | -2.70    |
| 4      | 28              | 117 ± 0.050a    | 115.60 ± 0.278c | -1.19    |
| 5      | 35              | 115.03 ± 0.057c | 113.60 ± 0.398  | -1.24    |

Treatment (T)

103.75\*\*

Period (P)

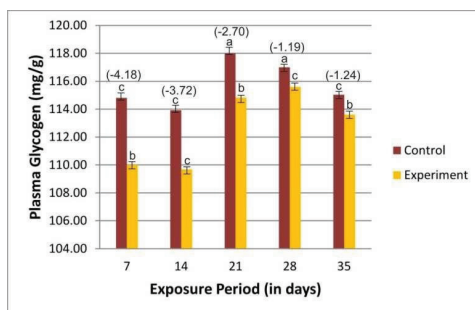
80.85\*\*

TXP

40.00\*\*

Values are mean ± S.E. of five individual observations. (+) Denotes percent increase over control. \*\* Significant at 5% level. Means in a column bearing same letter(s) are significantly different according to DMRT ( $P > 0.05$ )

**Fig. 13. Plasma glycogen content of *Cyprinus carpio* exposed to sublethal concentration of silver nanoparticles for 35 days. Error bars indicate the standard error of the mean. Bars bearing same letter(s) are significantly different according to DMRT ( $P > 0.05$ ). The numerical in the parenthesis indicates percent change.**



## DISCUSSION

Metabolism and biochemical processes of the organism are affected by biochemical profile changes, resulting from the effects of various pollutants, and they make it possible to study the mechanisms of the effects of substances. Adedeji (2010) reported under unsuitable environmental conditions (temperature, pH, oxygen concentration) or the presence of stressing factors, such as toxic chemicals (Barcellos *et al.*, 2004; Firat and Kargin, 2010) are indicated from alteration of blood chemistry. Biochemical and physiological methods of diagnosis constitute a promising approach to the problems of detecting the effect of toxic chemicals at possible stages (Larsson *et al.*, 1985). Detection stages of stress, suggestion of modes of action are tentatively the tools to explain the metabolic basis for conventional fishery statistics like growth. Appropriate biochemical parameters could be used effectively as sensitive indicators. Pollution may induce biochemical changes in fishes before the drastic cellular and systematic dysfunction manifest. Glucose is one of the most sensitive indices which cause stress to the organism. Earlier reports on the changes in the blood glucose in response to stress are contradictory showing both a rise and a fall. Its high concentrations in blood indicate that the fish under stress and intensively using energy reserves glycogen in liver and muscles (Firat and Kargin, 2010).

Glucose in serum is a major metabolite of carbohydrate metabolism (Artacho *et al.*, 2007; Zhou *et al.*, 2009) and blood glucose level have long been used as indicators of stress in fish (Osman *et al.*, 2010). Glycogenesis is the breakdown of glycogen (n) to glucose-6-phosphate and glycogen (n-1). Intensive glycogenolyses and synthesis of glucose from extra hepatic tissue, proteins and aminoacids (Vignesh *et al.*, 2013) in *Labeo rohita* and *Cyprinus carpio* which increases and decreases the glucose content in blood. Higher glucose content in blood decreases the activities of enzymes like profructokinase which in turn decreases the glycolysis capacity (Almeida *et al.*, 2001; Sobha *et al.*, 2007). The increase in serum glucose level in both *Onchornchus mykiss* and *Heteropneustesfossilis* exposed to cadmium and *Channa punctatus* and *Cyprinus carpio* exposed to mercury were linked with the decrease in glycogen content and reserves in the muscles and liver tissues (Sastry Rao, 1984; Richard *et al.*, 1988; Canli, 1996). Hence Blood glucose level has played a major role in the indication of stress. *Tilapia zilli* were exposed to cadmium, Ghazaly (1996) *Salmogairdneri* exposed to titanium nanoparticles. Acute decrease in the blood/plasma glucose level was noticed when *Channa punctatus* and *Cyprinus carpio* exposed to mercuric chloride (Ramesh, 1995).

When stress is induced, there is an increase in blood glucose level is reported. The release of adrenaline and nor- adrenaline to activate the secretion of catecholamine. Energy demand of a cell would be increased if there is a stress- induced increased cell metabolism. Catecholamine helps in the conversion of liver glycogen to blood glucose to supply the greater energy demand of the cells (Yeo and Kang, 2008). The plasma glucose level increases when rainbow trout *Salmo gairdneri* (Larsson and Haux, 1992) and *Notemigonus crysoleucas* (Gill and Pant, 1993; Benson *et al.*, 1997; Haux Larsson, 1994) are exposed to cadmium toxicity, which also causes reduction in the insulin secretion. According to the sample report blood glucose level was elevated in cat fish (*Heteropneustes fossilis*) when it was exposed to lead for 120 days leads to glucogenesis in muscle blood and liver.

A peculiar increase in blood glucose level was noticed in *Oreochromis aureus* when it was treated with Malathion. This was due to the stimulation of (GP) Glucose phosphorylase and decrease in oxygen consumption. Accumulation of Lead in *Oreochromis mossambicus* increased as a result from adaptation for survival under chronic mental stress. The increase serum glucose levels in both *Oncorhynchus mykiss* / *Heteropneustes fossilis* exposed to Cd and *Channa punctatus* / *Cyprinus carpio* exposed to Hg were linked with the decrease in glycogen reserves in the muscle and liver tissues as per the study by Sastry and Rao, 1984; Richard *et al.*, 1998. It was shown that serum glucose levels in fish were affected by many stress factors, including heavy metals. Studies have shown that glucose level in *Rainbow trout* was increased and affects the reproductive system; this was caused due to combined effects of pulp and paper effluent and biochemical effects from the suspended solid discharged from the mill. It was determined that cadmium increases the serum glucose levels and the activities of aspartate transaminase and alanine transaminase in *Notemigonus crysoleuca* and the activities of enzymes that play a role in the aminoacid catabolism like glutamate dehydrogenase, aminoacid oxylase and the xanthine oxylase in *Channa punctatus* (Sastry and Rao, 1984). Thus an increase in serum glucose levels without decrease in glycogen reserves in fish exposed to salts indicate the involvement of gluconeogenic events according to study by Sheridan and Mommsen, 1991. The progressive accumulation of the blood glucose suggests a stress induced, as a result of the incomplete metabolism of blood sugar due to impaired osmoregulation, as suggested by (Omoregie *et al.*, 1990). Also it was noted that serum glucose levels decreased with increasing concentrations of Cd in *Mugil cephalus*.

Decrease of glucose due to silver and silver nanoparticles exposure in fresh water fish *Cyprinus carpio* may be due to the decrease of metabolite in liver (Seok-Jeong, *et al.*, 2014).

Decrease in plasma glucose level was reported in the metabolic profile altered in two fishes the *Pacu* and *Tambacu* due to short term silver exposure (Moraes *et al.*, 2006). It is observed that in *Pacu*, there is a reduction in glucose level of liver and muscle, glycogen level increased in liver and decreased in muscle. In *Tambacu*, glucose levels were higher in the liver and plasma. The glycogen content of the liver and muscle turns rose in exposed fish. The increased behaviour is due to the metabolite named plasmatic pyruvate and reduction is due the absence of this metabolite in liver. *Tambacu's* liver provides comparatively higher glucose level in liver than *Pacu*, which were subjected to silver effect. And in silver nanoparticles glucose level decreased in fish *Cyprinus carpio* in the present study.

Carbohydrate metabolism is disturbed when animals are subjected to toxic stress and immediate energy demand of the body provided by glucose. When there is a physiological demand for energy, glucose may be oxidized completely to  $\text{CO}_2$ , water and energy is released. Study from (Ravichandran *et al.*, 1995) suggests that breakdown of glycogen (glycogenolysis) happens to cope with the high-energy demand due to stress caused by pollutants. Significant increase in glucose level in *Sarotherodon mossambica* has been noted due to exposure to cadmium (Cd), also in many fish species cadmium (Cd) induced hyperglycemia which has been reported along with depleted hepatic glycogen as per study by Bhattacharya *et al.*, 1987. Study from Ruparelia *et al.*, 1992 has shown that in order to meet the high energy demand of stressed animals, glycogen which is rapidly available for energy production may be rapidly catabolised resulting in huge or less of its reserve. Glycogen is the main source of energy for animals during normal metabolism and their amount in fish muscle exposed to chemical substances is an indicator for the health condition of the fish. They also reported that during unfavourable environmental situation the normal metabolism is affected, which in turn leads to alteration in the glycogen reserve of fish. The significant increase in glycogen content during sublethal lindane treatment may indicate an adaptation of fish to lindane toxicity or impairment in the carbohydrate metabolism as per Saravanan *et al.*, 2011. Sastry and Siddiqui (1984) reported that the muscle glycogen level was increased in quinalphos exposed fish indicating a decrease in the rate of glycogenolysis. In the present study, the significant decrease of glycogen is observed in acute and sublethal treatments indicate impairment in carbohydrate metabolism or a decrease in the rate of glycogenolysis. Generally, depletion of glycogen level under stress condition may be an indication of initial regulatory step resulting in an increase in intermediary metabolism. The reduction in glycogen content in *Cyprinus carpio* during acute and sublethal of silver and silver nanoparticles treatment indicates the utilization of stored glycogen to meet the high

energy requirement under the silver stress. The reduction of hepatic and muscular glycogen in *Silver catfish* exposed to high nitrate levels at different pH may indicate that stress generated by ammonia toxicity is accompanied by a rapid degradation of tissue glycogen. Studies from Vijayavel *et al.*, 2006 suggests that carbohydrates stored in muscles are the first nutrients used in response to stress conditions. Lower glycogen levels were observed in freshwater fishes exposed to high  $\text{NH}_3$  levels (1-4mg/L) as per study by Miron *et al.*, 2008. Depletion of glycogen may be due to direct utilization for energy generation, a demand caused by latex – induced hypoxia. Fish white muscles constitute more than 50% of the whole body mass, and anaerobic processes require glycogen for their action as suggested from study by Knox *et al.*, 1980. In the present study the significant decrease in glycogen level during sublethal treatment indicates that stress generated by silver toxicity is accompanied by a rapid degradation of glycogen. The alteration in glycogen content can be understood in two exclusive ways, the glucose can be converted to other body tissues, especially to that in the brain, or it can be converted into glycogen. The decrease in glycogen content is due to silver nanoparticles exposed fish might be due to metabolite named plasmatic pyruvate. Decreased glycogen level in muscle, in *Cyprinus carpio* exposed to silver may be due to the absence of metabolite in liver.

The engineered nanomaterial comprises of numerous different biochemical and physical forms and some of these materials includes carbon nanotubes, carbon spheres called fullerenes as per Zhu *et al.*, 2006 or nanoparticles made from metals as per Griffitt *et al.*, 2007 or metal oxides as per Federici *et al.*, 2007 or composites made of several metals as per King-Heiden *et al.*, 2009 and all these have adverse effects on fish as per the study from Smith *et al.*, 2007.

To study the aquatic toxicity of fish in glucose and glycogen level, fish species has been widely used as an indicator of pollutant and they strongly respond to stress conditions. In conclusion, this study provides evidence that silver nanoparticles reduce the ability of fish to extract oxygen from the water during progressive oxygen depletion, thereby increasing vulnerability to hypoxia. On the other hand, BMR, was unaffected by nanosilver exposure during normoxia. The action of nanosilver is most likely on the gill surfaces resulting in impaired gas exchange. Nano silver concentrations and the expected future use of nanosilver undoubtedly result in increased environmental concentrations. Toxicity from the silver is very less as compared to that of nanosilver.

# HISTOPATHOLOGY

## INTRODUCTION

The toxicity of nanoparticles depends on the shape, size, structure of the particle (Choi *et al.*, 2008). Widely used nanoparticles, such as silver nanoparticles, most likely enter the ecosystem change their densities or community populations and may produce a physiological response in many animals, possibly altering their fitness. Silver nanoparticles (AgNPs) are commonly used for solar energy absorption, spectrally selective coatings and acts as chemical catalysts especially for antimicrobial sterilization made them one of nonmaterial's in many applications (Pal, *et al.*, 2007). Open access literature regarding the toxicity of nanoparticles (Nps) is still emerging and gaps still exist in our knowledge of this area (Pal, Tak and Song, 2007).

Nanoparticles are small sized particles. More than 1.0 ppm of silver nanoparticles exposure causes toxic effect. Studies of nanoparticles in the field of toxicology are nanotoxicology. Nanotoxicology evaluates the harmful effects of Nps and their influence on the environment (Handy *et al.*, 2012). Biosystems may have serious ecological consequences and affect human and animal health due to the presence of Nps (Handy *et al.*, 2008). Chronic inhalation and consumption of Nps are the most dangerous implications (Moore, 2006). This implication of Nps leads to the histopathological changes and aggregation in the aquatic environment. The concentration of various nanoparticles for fish ranged from 100  $\mu\text{g L}^{-1}$  to 1  $\text{mg L}^{-1}$ , while the lethal concentration of nanoparticles reaches the milligrams per liter range. The concentrations of Nps in surface water range from nanograms per liter to low micrograms per liter which affects the organs like the gill, liver, kidney ( Handy *et al.* 2012).

Histopathological biomarkers have been recommended for field application, nonspecific response to several stressful stimuli and more often as a generalised experiment. This biomarker is updated as histopathology. Histopatology is an important component of several measures of fish health (The *et al.*, 1997). Changes in various tissue and organs in animal provide a rapid method to detect effects of irritants, especially chronic ones is the study of histopathology. In the polluted ecosystems the potential utility of biomarkers for monitoring both environmental quality and the health of organisms inhibiting has received increasing attention during the last few years. For monitoring the fish health and environmental pollution in natural water bodies the histopathological analysis has been tested

and proposed as an efficient and sensitive tool (The *et al.*, 1997; Stentiford *et al.*, 2003). In practical ecological risk assessment methodologies, an increasing amount of research is incorporating in histopathological biomarkers. Fish are used as an experimental model that has been widely used to evaluate the health of aquatic ecosystems of toxicology in pathology (Korkmaz, *et al.*, 2009).

In the aquatic environment, silver nanoparticles are utilized in an increasing amount of products, and discharge is inevitable. Fish gills are in direct contact with the ambient water, making them potential exposed and vulnerable to suspended silver nanoparticles. The present study investigates the effect of silver nanoparticles. The toxicity of silver nanoparticles is hypothesized due to disruption of membrane transport processes that disturb osmoregulation and affect the organs like gill, liver, and kidney and finally lead to cell death (Bilberg,*et al.*, 2010). In early development phases, taken up via food the toxicity of silver is high, when silver exposed to organisms (Luoma, 2008). Up take of silver ions ruptures the permeability of membrane and destroy cell wall which could cause cell lyses and silver ions also inhibit different important cycles (S, N and P) of nitrifying bacteria, disturb DNA transcription. Boyle, *et al.* 2013 investigated the mechanism of action of silver and silver nanoparticles to adult rainbow trout. He noticed that when an increase of salinity,  $\text{Cl}^-$ , in water seemed to be decrease the toxicity of nanoparticles. Here also suggests that the exposure of silver, silver nanoparticles along with nitrite, forms silver nitrite and  $\text{AgNO}_3$  give similar effects, occurrence of internal physiological disturbances, as exposure of acid conditions. They concluded that it was most likely because of  $\text{Ag}^+$  binds to the gill surface and interferes with  $\text{Na}^+/\text{Cl}^-$  uptake, which leads to decrease in plasma  $\text{Na}^+$  and  $\text{Cl}^-$  concentrations and fluid balance.

Half of the body surface area formed of the gill surface. In fish the external environment separated from the internal environment is only by a few microns of delicate gill epithelium and thus the bronchial function is very sensitive to environmental contamination. The gas and ion exchange between the organism and environment by the fish gills and this plays an important role (Witeska *et al.*, 2006; Oliva *et al.*, 2009). In the aquatic medium the principal sites of numerous toxic substances are the gill apparatus and the skin, which are continuously exposed in water. Within 30min of exposure, the toxic action of silver nanoparticles was relatively rapid with signs of stress. At higher absorbance silver nanoparticle concentrations ( $>80\mu\text{g L}^{-1}$ ), toxicity stress signs emerged, with increased respiratory rate. The surface respiration took place, where they ultimately lost equilibrium

and sank to the bottom. Few fishes displayed jerky movements and circular swimming just before they lost equilibrium. No signs of aggressive behaviour or changes in body colour were observed in any of the fishes in tanks.

Since gills are considered the most vulnerable organ to external pollutants owing to their direct contact with water, facilitating the lamellar epithelia penetration and ingress into blood circulation (Gernhofer *et al.*, 2001), several histological studies have been carried out in order to determine the physiological and morphological changes in gills (Mallat, 1985). Gill lesions as indicators of exposure to contaminants have previously been used in numerous laboratory and field studies around the world (Thophon *et al.*, 2003). Gills therefore, are potentially useful to monitor the health of fish (Lacroix *et al.*, 1993). The gills in fish can be a valuable model for assessing the effects of toxicants on cells and tissues are stated by Mallatt (1985). Structural alterations in gill of fish inhabiting polluted water have been studied by many workers (Bucher and hofer, 1993; Cengiz, 2006; Kelly and Janz, 2009). The changes in gills were observed in *Puntius gonionotus* (thickening of primary lamellar epithelium and clubbing of secondary lamella exposed to cadmium (Wangsongsak *et al.*, 2007). Krishnani *et al.* (2003) observed gill epithelial necrosis in *Lates Calcarifer* exposed to copper. Das *et al.*, (2004) in *Labeo rohita* observed gill epithelial necrosis due to nitrite exposure. Mishra and Mohanty (2008) observed the changes in gill of *Channa punctatus* like epithelial necrosis and desquamation and aneurism exposed to chromium. Nile tilapia exposed to cypermethrin treatments shows lifting of epithelia, oedema and hypertrophy of epithelial cells (Korkmaz *et al.*, 2009). Cengiz (2006) reported desquamation and necrosis in the gill of *Cyprinus carpio* exposed to deltamethrin. Silver nanoparticles exposed to fish *Cyprinus carpio* and *Siberian sturgeon* (Teresa Ostaszewska *et al.*, 2016)

Liver plays an important role in vital functions. The fundamental metabolism starts in the liver. Moreover, it is the major organ of accumulation, biotransformation, and excretion of contaminants in fish (Wu *et al.*, 2008). Histopathology of fish liver is a monitoring tool that can provide an assessment of the effects of environmental stressors on fish populations, and it is proposed to be one of the most reliable organ (Hinton and Lauren, 1990; Fernandes *et al.*, 2008). Brusle *et al.*, (1996) stated that liver histology serve as a model for studying the interactions structure testify the development of cirrhosis or lipid degeneration of liver. Aquatic organisms especially fishes observe silver nanoparticles in water and it is accumulated in the liver. Pathological findings in the liver of fish exposed to silver nanoparticles include lipid peroxidation with enhanced content of ceroid/lipofuscin, necrosis,

increased number and size of macrophage aggregates or alterations in cytoplasmic organization (Oliveria-Ribeiro *et al.*, 2002).

Similar results have also been reported by many authors (Venkateshwara Rao, 2006; Korkmaz *et al.*, 2009). Bhattacharya *et al.*, (1987) in *Clarias batrachus* and Konar (1970) in *Labeo rohita* observed histopathological changes in liver like liver cord disarray, vacuolization and necrosis of liver when treated with endrin, industrial pollutants and sevin, respectively. Pane *et al.* (2005) observed a significant contraction of the lamellar vasculature following tributyltin compounds and nickel. Histological changes in liver of *Oreochromis mossambicus* exposed to cadmium and zinc shows hyalinization, hepatic vacillation cellular swelling and congestion of blood vessels. Kahru and Dubourguier (2010) classified the Nps of silver and zinc oxide (ZnO) as “extremely toxic,” while other Nps are “toxic” or “harmful.” Silver nanoparticles (AgNPs) are commonly used in various industries (food, textile, paint, or electronics), in different kinds of antibacterial layers. AgNps may also generate the production of oxidants which are responsible for the destruction of the bacteria cell membrane (Kim *et al.* 2007). AgNPs also affect the protein membranes (OmpA, OmpC, OmpF) causing changes in their structure and functioning, and may affect heat shock proteins (IbpA, IbpB). They also cause changes in 30s ribosomal subunit. It was observed that AgNps have proapoptotic and anti-inflammation function as well and damage the liver cells (Choi *et al.*, 2010).

Kidney is the predominant organ involved in the regulation of extra cellular volume and control of electrolyte and acid-base balance and major site of foundation of hormones that influence systemic metabolic functions (Hook, 1980). In fish, as in higher vertebrates, the kidney performs an important function related to electrolyte and water balance and the maintenance of stable internal environment. The kidney excretes nitrogen-containing waste products from the metabolism such as ammonia, urea and creatinine. In teleosts, the kidney, together with the gills and intestine, are responsible for excretion and the maintenance of the homeostasis of the body fluids (Hinton *et al.*, 1992) and besides producing urine, act as an excretory route for the metabolites of a variety of xenobiotics to which the fish may be exposed (Hinton *et al.*, 1992). Since a large volume of blood flows through the kidney, lesions found in this organ can be useful as signs of environmental pollution (Hinton and Lauren, 1990).

Kidney damage the glomerulus, shrinkage of Bowman's capsule, vacillation formed around the capsule, enlargement of tubular lumen and distal and proximal tubules were observed in *Chanma Punctatus* exposed to tannery effluents (Mohanta *et al.*, 2010). Following exposure of fish to toxic agents such as pesticides, tissue alterations have been found at the level of the tubular epithelium and glomerulus (The *et al.*, 1997). Shrinkage of glomerular tuft, dilation of the lumina of the kidney tubules and necrosis of tubules has also been reported in fish exposed to different pollutants especially in silver nanoparticles (Venkateswara Rao, 2006). In some species of fish kidneys severely damages by the effects of pollutants, depends on the sensitivity of the species to the substances released in to the environment. Kidney damage hypertrophy and vacillation of the epithelial cells lining, proximal convoluted tubules, shrinkages in the glomeruli resulting in dilation of Bowman's space, necrotic changes in *Liza paretis* exposed to lead. Abdul- Tawwab *et al.* (2007) observed kidney damage in *Nile tilapia, Oreochromis niloticus(L)* exposed to copper and nanoparticles.

Silver nanoparticles products are characterized by high percentage of silver metallic form. This is important because ionic silver becomes silver chloride in the stomach, bloodstream and in kidney. Recent toxicity studies with vertebrates have indicated that silver may still be adversely affecting the health of kidney in fish. Silver causes lethal damage to hepatocytes cells and glomeruli cells and finally it leads to cell death (Baldi *et al.* 2002). Concentration of silver has been found to be less than 5ng L<sup>-1</sup> in three Connecticut undeveloped headwaters and between 25 and 100ng L<sup>-1</sup> in rivers from industrialised and urban areas (Rozan *et al.*, 1995).). Future discharge of nanosilver to the aquatic environment will undoubtedly increase seriously due to the expected extensive use of nanosilver and causes damage in gill, liver and kidney. Titanium dioxide nanoparticles caused respiratory disturbance in rainbow trout (*Oncorhynchus mykiss*) (Federici *et al.*, 2007), single walled carbon nanotubes elevated ventilation rate in rainbow trout (Smith *et al.*, 2007) and silver nanoparticles exposure induced surface ventilation and damage the gill, liver and kidney in *Cyprinus carpio* (own observations). It has also been suggested that copper nanoparticles act on the gills in *zebrafish* (Griffitt *et al.*, 2007).

Uptake of silver nanoparticles by aquatic organism is likely through membrane, epithelia, gills and surface. Moore (2006) states that at cellular level, most intakes of nanoparticles will occur via endocytosis. Silver ions have been found to be toxic to several organisms such (Ratte, 1999; Moore 2006; Choi *et al.*, 2008; Navarro *et al.*, 2008). The results

indicate that AgNPs exposure negatively influenced survival; body length and mass; and morphology and physiology of the epidermis, gills, and liver in fish. Fish exposed to AgNPs showed similar pathological changes: irregular structure and pyknotic nuclei of epidermis, aplasia and/or fusion of lamellae, telangiectasis, epithelial necrosis and lifting of the gills, dilation of sinusoidal space, overfilled blood vessels, and pyknotic nuclei of the liver. The tissues and organs of fish are affected by silver iron exposure leads to a slight increase in mortality. As mortality is the most obvious sign of progression of serious pollutant at the organism level, the impairment role of survival due to AgNPs exposure may be considered a consequence of a serious progression of population mortality. In recent years, silver and silver nanoparticles (AgNPs) are used widely in applications, animal and eventually human exposure to AgNPs has been increased. It has been confirmed that AgNPs are translated into blood circulation and accumulated in organs like gill, liver and kidney and causes organotoxicity, and histopathological test were examined and finally death occurs.

## **MATERIALS AND METHODS**

### **HISTOLOGICAL STUDIES**

The methods of pears (1968), Roberts (1978) and Humason (1979) were followed for histopathological studies of gill, liver and kidney.

### **FIXATION**

The gills, liver and kidney were dissected out. The fixative, prevent bacterial decay and autolysis were cut bits of 1 to 2 cm in diameter. These tissues were immediately put in Bouin's fluid for 24h. To avoid post – mortem changes and shrinkage during further process like dehydration, embedding and sectioning.

### **WASHING**

After fixation the tissues were taken and the excessive fixation was removed by transferring the tissues to 50% alcohol (to prevent interference with subsequent process).

### **DEHYDRATION**

In dehydration process the tissues were put in to alcohol series like 30%, 50%, 70%, 90% and absolute alcohol and duration of 30 min. was given in each alcohol series. A

minimum of 2 or 3 changes were given in absolute alcohol at an interval of 30 minutes for ensure complete removal of water from the tissues.

## **CLEARING**

During clearing alcohol was replaced from the tissues by using a clearing agent xylene. The tissues were kept in xylene for about 30 min to 1h until they became transparent.

## **INFILTRATION**

During infiltration the tissues were kept in a paraffin embedding bath (metal cups filled with paraffin at 58<sup>0</sup>-60<sup>0</sup>C). Then the tissues were directly transferred from xylene to molten paraffin. A minimum of three changes were given in paraffin wax with 30 minutes duration in each.

## **EMBEDDING**

For embedding, L' blocks were filled with molten paraffin wax. Then the tissues were placed with proper orientation. After embedding process, the blocks were kept in water overnight to ensure complete solidification. Finally, blocks were removed for sectioning.

## **SECTIONING**

Using a rotary microtome, tissues were cut with 7  $\mu$  thickness. Then the sections were spread on a glass slide using egg albumin as an adhesive. After complete spreading, the sections were placed in an oven overnight at 37<sup>0</sup>C and the sections were taken for staining.

## **STAINING**

During staining paraffin wax from the sections was removed by dewaxing using xylene. Then the sections were hydrated when it immersing in descending grades of alcohol (absolute alcohol, 90%, 70%, 50%, 30%) for about 1-2 min. in each. Then, the sections were stained in Heidenhain's iron heamatoxylin stain for 2 to 5 min, washed in tap water until the sections became bluish black in colour and then stained in 1% Eosin.

For restaining, 1% Iron alum was used. The sections were dehydrated through ascending grades of alcohol (30%, 50%, 70%, 90%, absolute alcohol) for about 5sec. in each. Subsequently, the sections were cleared in xylene and mounted permanently with cover glass

using DPX mountant. The histological results of gills, liver and kidney of fish from control and nitrite treated were given as photomicrographs in appropriate places in the text.

### **Nanoparticles (AgNPs ) used in the experiment**

Nanosilver (cat. no 576832, Sigma Aldrich, UK) preparation was based on the manufacturer's specification for silver nanopowder of particle size <100 nm, surface area  $5.0 \text{ m}^2 \text{ g}^{-1}$ , density  $10.49 \text{ g cm}^{-3}$ , and purity of 99.5 %. Silver nanoparticles  $50 \text{ mg L}^{-1}$  were dispersed by sonication in Milli-Q water for 30 min (sonicator, 250 W, 40 kHz, 25 °C; Ultrason U-507 Ultrason, Poland). After sonication the solutions were filtered through the 200-nm nylon membrane filter (Whatman®, UK). The distribution and size of nanoparticles were inspected by transmission electron microscopy (TEM) using a JEOL JEM-1220 TE microscope at 80 KeV (JEOL Ltd., Japan), with Morada 11 megapixel camera (Olympus Corporation, Japan). The Zeta potential of the nanoparticles was measured by electrophoretic light-scattering method, using Zetasizer Nano-ZS90 (Malvern, Worcestershire, UK). Each sample was measured after 120 s of stabilization at 25 °C, pH 8.6, in 20 replicates.

## **RESULT**

### **HISTOLOGICAL OBSERVATION IN GILL**

In control fish, gill filaments were seen and secondary lamellae were lined along both sides of the gill filament. The surface of the control lamellae was covered with epithelial cells running parallel along the surface. This epithelium consists of different cell types including pavement (epithelial), mucous (goblet), and chloride Ph. m. 1.

Histopathological structure of *Cyprinus carpio* exposed to acute concentration of silver nanoparticles was shown in Ph. m.2. During acute treatment, fish showed increased mucous secretion, hyperplasia of the epithelial cells, hyperplasia and fusion of secondary lamella.

Ph.m. 3 shows the gill morphological structure of *Cyprinus carpio* when exposed to sublethal concentration of silver nanoparticles for 7 days. During the exposure period, the primary gill lamellae swollen due to hyperplasia of epithelial cells and fusion of the marginal areas of adjacent lamellae were noticed. At the end of 14<sup>th</sup> day, mucous cells were located in the secondary lamellae and distally in the primary lamellae, while chloride cells were only observed on the basement of the secondary gill lamellae (Ph. m. 4).

Ph. m. 5 and 6 shows the gill morphological structure of *Cyprinus carpio* when exposed to sublethal concentration (21<sup>st</sup> and 28<sup>th</sup> day) of silver nanoparticles. During the above treatment, hypertrophied epithelial, lamellar hyperplasia in the middle part of secondary lamella, epithelial lifting, aneurysms lifting, lamellar fusion, and bulging was detected. The observed changes were severe as the exposure period extended showing a complete rupture of gill at the end of 35<sup>th</sup> day (Ph.m 7).

## GILL TISSUES

Gill tissues of *Cyprinus carpio* exposed to 1.0 ppm of silver nanoparticles for 24h (acute): GF :Gill filament, LF : lamellar fusion, ECN : Epithelial cell necrosis H&E, X 100..

Gill tissues of *Cyprinus carpio* exposed to 0.01 (sublethal) ppm of silver nanoparticles for 7, 14, 21, 28 and 35 days. The changes in the gills were observed in H&E, X 100.

Exposure of silver nanoparticles gill tissues of 7<sup>th</sup> day the changes observed were EL : Epithelial lifting, HP : Gill exhibiting hyperplasia, EN : Epithelial necrosis H&E, X 100.

Exposure of silver nanoparticles in gill tissues of 14<sup>th</sup> day the changes observed were H : Hypertrophy of lamellar epithelium, ECN : Epithelial cell necrosis, LF : lamellar fusion H&E, X 100.

Exposure of silver nanoparticles in gill tissues of 21<sup>th</sup> day the changes observed were GF : Gill filament, GL : Gill lamella, H : Hypertrophy of lamellar epithelium H&E, X 100.

Exposure of silver nanoparticles in gill tissues of 28<sup>th</sup> day the changes observed were EN : Epithelial necrosis, GC : Goblet cell, GF : Gill filament H&E, X 100.

Exposure of silver nanoparticles in gill tissues of 35<sup>th</sup> day the changes observed were GF : Gill filament, EN : Epithelial necrosis, GC : Goblet cell H&E, X 100.

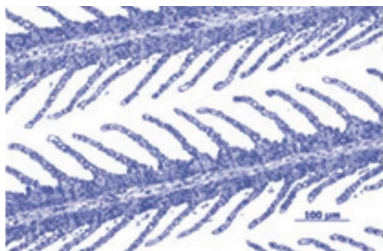


Fig. 1

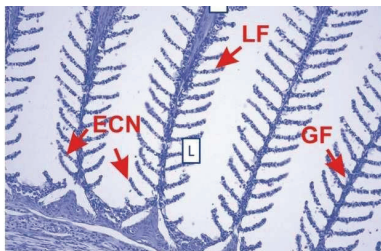


Fig. 2

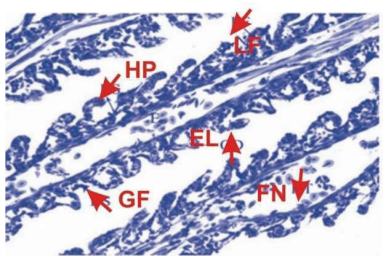


Fig. 3

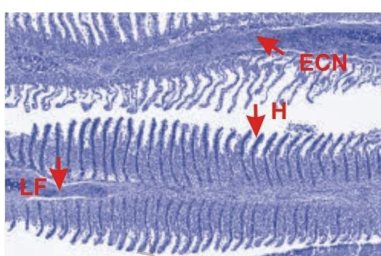


Fig. 4

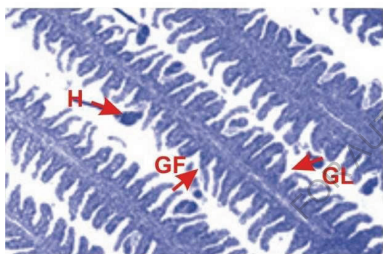


Fig. 5

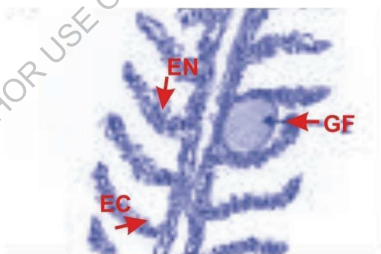


Fig. 6

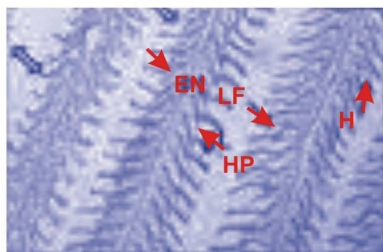


Fig. 7

## HISTOPATHOLOGICAL OBSERVATIONS IN LIVER

Morphological structure of liver of *Cyprinus carpio* was presented in Phm 8.

In the control fish, the liver showed hepatic cells located with central nucleus and homogenous cytoplasm. During acute treatment the liver exhibit disintegration of parenchymal cells, extensive vacillation of parenchymal hepatocytes, degenerative cell organelles and lumen with necrotic cells (Ph. m. 9).

During sublethal treatment liver showed enlargements of hepatocytes, disintegration of cell boundaries of intercellular spaces, nuclear degeneration with pyknotic nucleus etc., were noticed at the end of 7<sup>th</sup> day (Ph. m. 10). At the end of 14<sup>th</sup> day, disintegration of hepatocytes and necrosis of parenchymatous tissue noticed. The cell undergoes necrosis or apoptosis. It is followed by Karyorrhexis or fragmentation of nucleus (Ph. m. 11). The above said structural changes were severe at the end of 21<sup>st</sup> and 28<sup>th</sup>, exposure period (Ph. m. 13). Complete ruptures of hepatic cells, vascular degeneration, intercellular edema etc., were observed at the end of treatment period (Ph. m. 14).

### LIVER TISSUES

Liver tissue of *Cyprinus carpio* exposed to 1.0 pp of silver nanoparticles for 24h (acute): VD : Vascular degeneration, N : Necrotic hepatocytes H&E, X 100.

Liver tissue of *Cyprinus carpio* exposed to 0.01 ppm of silver nanoparticles for 7 days (sublethal): ND : Nuclear degeneration, CE : Cellular edema H&E, X 100.

Liver tissue of *Cyprinus carpio* exposed to 0.01 ppm of silver nanoparticles for 14 days (sublethal): NP : Nuclear pycnosis, NC : Necrotic cells H&E, X 100.

Liver tissue of *Cyprinus carpio* exposed to 0.01 ppm of silver nanoparticles for 21days (sublethal): IE : Intercellular edema, NP : Nuclear pycnosis H&E, X 100.

Liver tissue of *Cyprinus carpio* exposed to 0.01 ppm of silver nanoparticles for 28days (sublethal): VD : Vascular degeneration, IE : Intercellular edema, PN : Pyknotic nucleus H&E, X 100.

Liver tissue of *Cyprinus carpio* exposed to 0.01 ppm of silver nanoparticles for 35 days (sublethal): ND : Nuclear degeneration, VD : Vascular degeneration, N : Necrotic hepatocytes H&E, X 100.

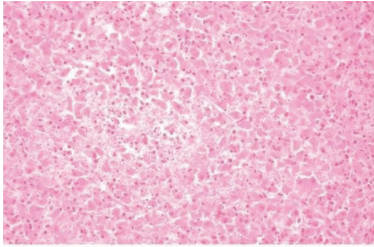


Fig. 8

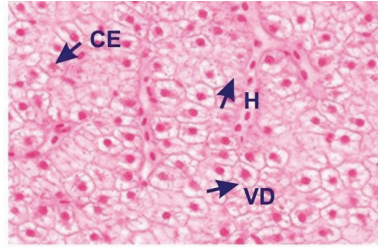


Fig. 9

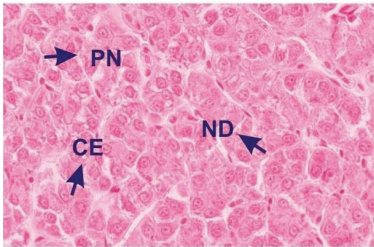


Fig. 10

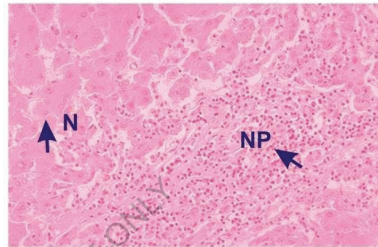


Fig. 11

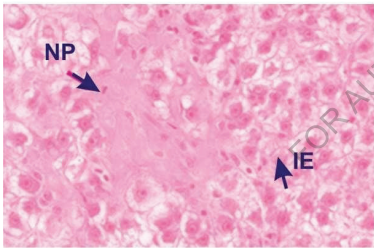


Fig. 12

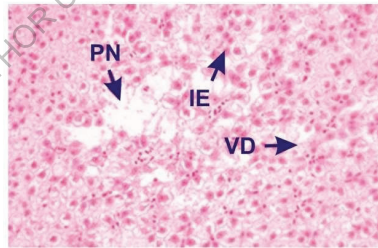


Fig. 13

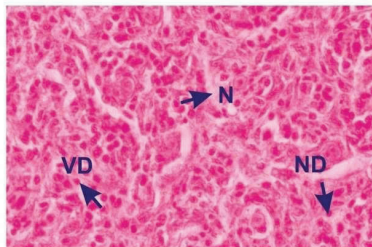


Fig. 14

## HISTOPATHOLOGICAL OBSERVATIONS IN KIDNEY

Ph.m.15 presents the normal structure of kidney of *Cyprinus carpio*. In control fish, the histology showed normal disposition of renal tubules, glomeruli and interspersed lymphoid tissue and nephric tubules were noticed (Ph. m. 16).

In sublethal treatment, the tubular epithelial cells of the kidney exhibited pyknotic and karyorrhectic nuclei and a disrupted luminal surface at the end of 7<sup>th</sup> day (Ph. m. 17). Further the kidney showed tubular necrosis, hyaline droplets degeneration, and renal tubular separation as the exposure period extended (14<sup>th</sup> and 21<sup>st</sup> day) (Ph. m. 18, 19). At the end of 28<sup>th</sup> day, kidney showed marked contraction of glomeruli, dissolution of lymphoid tissue and its replacement by blood forming sinuses, leading to formation of cellular debris, tubular necrosis and renal tubular separation (Ph. m. 20). At the end of the exposure period (35<sup>th</sup> day), kidney appeared spongy and friable; the glomeruli were conspicuously shrunken and collapsed, pyknotic nucleus in tubules, coagulative necrosis, nuclear pycnosis, renal tubular separation and hyaline droplets degeneration (Ph.m 21).

### KIDNEY TISSUES

Kidney tissue of *Cyprinus carpio* exposed to 1 ppm of silver nanoparticles for 24hours (acute): GN : Globular necrosis, TN : Tubular necrosis H&E, X 100.

Kidney tissue of *Cyprinus carpio* exposed to 0.01 ppm of silver nanoparticles of 7<sup>th</sup> day (sublethal): RTS : Renal tubular separation, HDD : Hyaline droplets degeneration H&E, X 100.

Kidney tissue of *Cyprinus carpio* exposed to 0.01 ppm of silver nanoparticles of 14<sup>th</sup> day (sublethal): TN : Tubular necrosis, RTS : Renal tubular separation H&E, X 100.

Kidney tissue of *Cyprinus carpio* exposed to 0.01 ppm of silver nanoparticles of 21<sup>st</sup>day (sublethal): CN : Coagulative necrosis, PN : Pyknotic nucleus in tubules H&E, X 100.

Kidney tissue of *Cyprinus carpio* exposed to 0.01 ppm of silver nanoparticles of 28<sup>th</sup>day (sublethal): TN : Tubular necrosis, RTS : Renal tubular separation, NP : Nuclear pycnosis H&E, X 100.

Kidney tissue of *Cyprinus carpio* exposed to 0.01 ppm of silver nanoparticles of 35<sup>th</sup> day (sublethal): HDD : Hyaline droplets degeneration, RTS : Renal tubular separation, TN : Tubular necrosis. H&E, X 100.

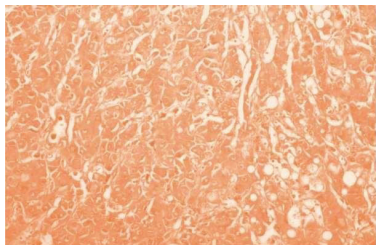


Fig. 15

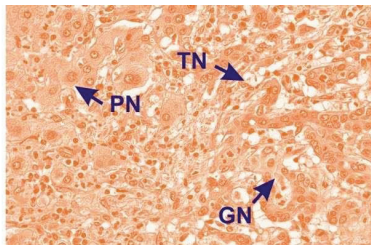


Fig. 16

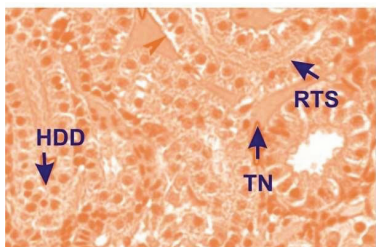


Fig. 17

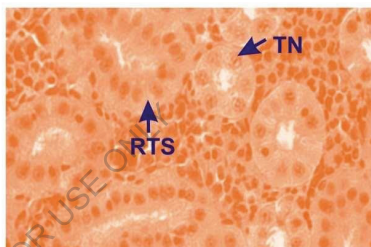


Fig. 18

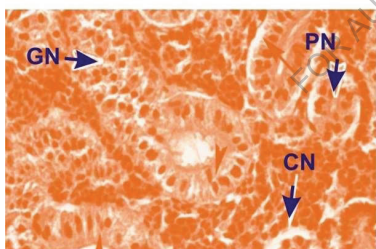


Fig. 19

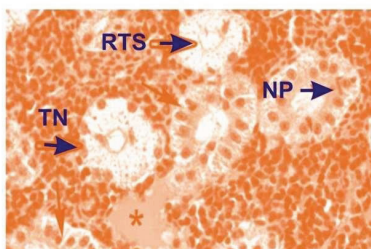


Fig. 20

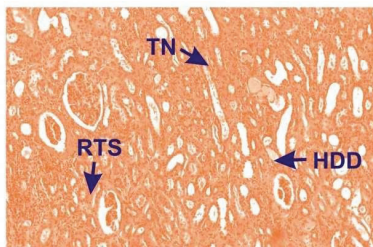


Fig. 21

## DISCUSSION

Nanoparticles adversely affect fish growth and survival. Histological analyses revealed histopathological lesions caused by AgNPs. These results indicate the toxic action of silver nanoparticles to fishes accompanied by low survival. The most severe alterations were observed in fish exposed to  $1.5 \text{ mg L}^{-1}$  and  $1.75 \text{ mg L}^{-1}$  of silver nanoparticles in gill, liver and kidney of freshwater fish. In the epithelial layer epidermal lesions were found, and their frequency and severity increased with the increase in nanoparticle concentrations. Gills are in intimate contact with water and with the external environment, represent a thin and extensive surface (up to 90% of the total body surface) and it is the first target of waterborne pollutants (Poleksic Tutundzic, 1994; Fernandes and Mazon, 2003).

The most commonly observed alterations in gill is thenuclear pyknosis in external layer, irregular structure of organs, contraction of cytoplasm resulting in intercellular spaces, epidermal epithelium and increase in goblet cells and increased mucus secretion. The increase in number of goblet cells and increased mucus secretion are considered the first protective reaction to toxic agents and may temporarily reduce toxic impact (Handy and Maunder 2009). One of the great advantages of using histopathological biomarkers in environmental monitoring is that, this category of biomarkers allows examining specific target organs, including gill, kidney and liver that are responsible for vital functions, such as respiration, excretion, accumulation and biotransformation of xenobiotics in the fish (Gernhofer *et al.*, 2001). Oilveria Riberio *et al.*, (2002) reported that specific lesions occurring in organs of fish exposed to toxic substances under laboratory conditions help to identify biomarkers which are very sensitive, and can reveal toxicant etiology consistent with the type of environmental contamination (Handy *et al.*, 2002). Similar lesions in the epidermal epithelium of sterlet (*Acipenser ruthenus* L.) exposed to nanoparticles were reported by Poleksic *et al.* (2010).

In unfavourable environmental conditions gills are a well known target organs in fish, being the first to react (Benli *et al.*, 2008). Fish gills are well supplied with blood to facilitate gas exchange and have a large surface area covered by a thin layer of epithelial cells. Hence, gills are often most affected by exposure to toxicant-induced structural changes (Mallatt, 1985). In the present study by examining through the light microscope control individuals did not show any histopathological changes in the tissues. However, during acute and sublethal treatment of silver nanoparticles in the fish *Cyprinus carpio* shows histological alterations

such as rupture and lifting of the lamellar epithelium, hyperplasia of the lamellar epithelial cells, haemorrhage are showed by acute and sublethal treatment of silver nanoparticles. During acute treatment the hyperplasia was more severe, resulting in the fusion of some secondary lamellae. Further, alterations such as hypertrophy of epithelial cells, lamellar aneurysms and lamellar disorganization were also observed. Similar observation was made by many authors (Benli *et al.*, 2008). During prolonged exposure edema and hyperplasia of the respiratory epithelium of secondary lamellae was observed. In the present investigation during sublethal treatment separation of respiratory epithelium, alteration of pavement cells lifting and swelling of the lamellar epithelium and lesions were observed. The above structural alterations were severe when extended. Most differences between control and sublethal toxicant exposed fish were observed in the gills which include lamellar deformations; fusion of secondary lamellae and telangiectas is compared to control group.

Toxicant accumulated in the gills and damages the gill surfaces. This results in increased ventilation volume and decreased gill oxygen uptake efficiency, irritation due to elevated mucous secretion. In rainbow trout *Oncorhynchus mykiss* exposed to petroleum residues accumulated in gills, Engelhardt *et al.*, (1981) observed epithelial lifting and lamellar fusion. In fish *Astyanax altiparanae* collected from contaminated water Winkaler *et al.*, (2001) found anomalies such as hyperplasia, hypertrophy, dilation of the marginal channel and aneurysms in neotropical. In *Silver catfish* gills exposed to ammonia level above 96h- LC50, Miron *et al.* (2008) reported that edema and fusion of the secondary lamellae occurs. They also suggested that all these lesions may reduce gill functional surface for gaseous exchange, impairing respiratory function. Some alterations in blood vessels may also occur; when fishes suffer a more severe type of stress examined by Hinton and Laurn (1990) and he reported that most part of the gill lesions caused by sublethal exposure affects lamellar epithelium. In this case, damaged pillar cells can results in blood congestion or an aneurysm, an increased blood inside the lamellae, causing dilation of the marginal channel (Takashima and Hibiya, 1995). The formation of an eneursym is related to the rupture of the pillar cells (Martinez *et al.*, 2004) due to direct effect of contaminants on these cells. This is a severe type of lesion, but more difficult than the epithelial changes (Poleksic and Mitrovic – Tutundzic, 1994).

Uptake of silver and silver nanoparticles has been demonstrated in fish gills exposed through the water (Power, *et al.*, 2011), but possibly the uptake rate is affected by the

increased water flow passing the gills during hypoxia. Handy *et al.*, 2008 observed silver nanoparticles are taken up by adult fish is at present uncertain, but the pathway is probably different from that of silver ions due to the size of nanosilver. Dark field of microscopy of several organs of *Zebrafish* exposed to nanosilver reveal nanosilver deposits in very low. In the experimental case study of Braydich-Stolle *et al.* (2005), the cytotoxicity effects of silver nanoparticles on fish gills were investigated. In this study, the effect on cell morphology of gill, liver and kidney were taken into consideration. It was concluded that silver nanoparticles at 10 µg /ml, and above concentration showed dramatic changes like necrosis and apoptosis of cells. The concentrations of nanosilver used in the present study in *Cyprinus carpio* are higher than the estimated concentrations of nanosilver in the aquatic environment Prolonged usage of this silver protein as a medication caused discoloration in the fingernails, ocular agyrosis and damages the kidney cells.

Histopathological lesions occurs in in freshwater fish *Cyprinus carpio* exposed to AgNPs (Wu and Zhou 2013), CuNPs (Al-Bairuty *et al.*, 2013), TiO<sub>2</sub>NPs (Boyle *et al.*, 2013), and other aquatic pollutants and the changes such as hyperplasia, epithelial hypertrophy, lifting and telangiectasia were observed and shortening and fusion of gill lamellae and epithelial hyperplasia reduce contact of gills with water which results in reduced gas and ion exchange. Bilberg *et al.* (2010) reported respiratory disturbances and impaired tolerance to hypoxia in *Eurasian perch* after 24 h of nanosilver exposure. Hypoxic status induced by histopathological lesions was observed in *Japanese medaka* (Wu and Zhou 2013), and according to these authors it might have resulted in oxidative stress. In the present study nanoparticles of silver causes dilatation of lamellar blood vessels and aggregation of blood cells. According to Martinez *et al.* (2004), such changes may indicate damage of pillar cells and blood vessels which result in an increase of lamellar blood flow. *Siberian sturgeon* exposed to AgNPs and CuNPs showed also telangiectasia, epithelial detachment, and epithelial lifting. Epithelial lifting and detachment in secondary lamellae were also observed in *Japanese medaka* exposed to AgNPs (Wu and Zhou 2013). According to various authors, epithelial lifting usually results from edema of the secondary lamellae (Fanta *et al.*, 2003). Edema in gills of fish exposed to nanoparticles.

Nanoparticles inhibit ion transport by the bronchial Na<sup>+</sup> and K<sup>+</sup>-ATPase, which results in osmotic imbalance (Shaw et al. 2012). Bronchial lesions in *Siberian sturgeon* and *Cyprinus carpio* caused by AgNPs and CuNPs resulted in cell degeneration and epithelial

necrosis, similarly as in Atlantic salmon (*Salmo salar*) exposed to AgNPs the same results were observed (Farman *et al.*, 2012). Since, in general, these results serve as a barrier to the entrance of contaminants (Mallatt, 1985; Poleksic and Mitrovic Tutundzic 1994; Fermands and Mazon, 2003). Diffusion occurs in the epithelial area and thus reduces the absorption of pollutants into the blood; epithelial hyperplasia is considered as a protective response to toxic, irritants and environmental stressors (Mallatt, 1985). Lamellar aneurism, on the other hand, represents a lesion that can result from the rupture of pillar cells, and thus corresponds to the deleterious effect of xenobiotics on bronchial tissue (Martinez *et al.*, 2004) and alterations occur in these cells.

Alterations like epithelial lifting, epithelial cells, besides partial fusion of some secondary lamellae are the defense mechanisms of gills. According to Mallat (1985) such alterations are non-specific and may be induced by different types of contaminant. He concludes that the most common gill lesions induced by toxic substances and other chemicals lead to necrosis, hyperplasia, hypertrophy and rupture of gill tissues, lamellar fusion, hypersecretion and proliferation of mucous cells, alterations in chloride cells and vacuolarisation. Earlier authors divided the commonly reported gill lesions into two groups; (i) the direct deleterious effects of the irritants and (ii) the defense responses of the fish. In the present investigation the exposure of silver nanoparticles to fish *Cyprinus carpio* the changes observed were epithelial necrosis and desquamation of the gill epithelium are direct responses induced by the toxicant. The lifting of the epithelium increases the distance through which the toxicant has to travel to reach the blood stream. Lamellar fusion could be protective that diminishes the amount of vulnerable gill surface area. Gill hyperplasia might serve as a defensive mechanism leading to a decrease in the respiratory surface and an increase in the toxicant blood diffusion distance.

Liver being the main organ of various key metabolic pathways, toxic effects of chemicals usually appear primarily in the liver (Benli *et al.*, 2008). The organ most associated with detoxification and biotransformation process is the liver, due to its function, position and blood supply. It is also one of the organs most affected by metal nanoparticles in the water (Rodrigues and Fanta, 1998). Stentiford *et al.* (2003) observed that toxic pathology in liver lesions in fish species are effective biological markers of heavy metal exposures in silver nanoparticles. In the present study during acute and sublethal treatment the liver of *Cyprinus carpio* showed various alterations such as irregular shaped nuclei, nuclear hypertrophy, nuclear vacillation and vacuolization of the hepatocytes. Similarly, cytoplasm and nuclear

degeneration were also noticed. Anomalies, such as irregular shaped hepatocytes cytoplasmic vacillation and nucleus in a lateral position, close to cell membrane, were described in the *Siluriform corydoras* and *Paleatus* contaminated by organophosphates pesticides (Fanta *et al.*, 2003). Histological response in zinc exposed specimens with the most prevalent histological characteristics identified being hyalinization of hepatocytes, increased vacuolation associated with lipid accumulation, congestion of blood vessels, and cellular swelling (Hinton and Lauren 1990). The hexavalent chromium exposure exhibited marked degenerative changes in the liver tissues (Mishra and Mohanty, 2008). The presence of necrosis is in fact one of the most visible damages in tissues affected by nanoparticles (Rabitto *et al.*, 2005).

The hepatic histopathological lesions are often evaluated in toxicological studies and used as markers of environmental pollution due to nanoparticles (Dabrowska *et al.*, 2012). The liver shows a high potential of enzymatic degradation of toxic compounds, but it may be itself adversely affected by their high concentrations (Bruslé *et al.*, 1996). Hepatic histopathological changes in freshwater fish *Cyprinus carpio* exposed to various nanoparticles like AgNps, CuNps, TiO<sub>2</sub> nanoparticles were reported (Rabitto *et al.*, 2005). Severity of hepatic histopathological alterations in *Sturgeon* increased with the increase in nanoparticle concentrations. The livers of fish exposed to 0.1 and 0.5 mg L<sup>-1</sup> of AgNPs and to 0.01 and 0.05 mg L<sup>-1</sup> of CuNPs showed hepatocyte vacillation and increase in size compared to the control. Similar changes were observed by Hao *et al.* (2009) in the liver of carp (*Cyprinus carpio*) exposed to TiO<sub>2</sub>NPs in the liver of rainbow trout exposed to CuNPs. Abnormal accumulation of triglycerides and other neutral lipids may cause formation of vacuoles in hepatocytes and can be accompanied by pathological lesions such as necrosis (Kelly and Janz 2009). Vacillation of hepatocytes and the presence of pyknotic nuclei are indicative of the early stages of necrosis (Hao *et al.*, 2009) and found dilation of sinusoid space in the liver of Mozambique tilapia (*Oreochromis mossambicus*) treated with AgNPs.

Similar alterations were observed in *Sturgeons* exposed to 0.5 and 1.5 mg L<sup>-1</sup> of AgNPs and to 0.05 and 0.15 mg L<sup>-1</sup> of CuNPs. An increase in sinusoid diameter results from the reduction of hepatocyte size. Hepatocytes of fish exposed to the highest concentrations of both nanoparticles decreased in size and showed karyolysis. Such changes indicate progressive hepatocyte apoptosis and degeneration of hepatic parenchyma caused by the toxic action of nanoparticles (Choi *et al.*, 2010). On the contrary, exposure of rainbow trout to CuNPs caused a decrease in hepatic sinusoid space which indicates redirection of the blood flow to other organs. Hyaline degeneration (storage of the peptides from degraded cells) in

the liver, kidney, and gills induced by nanoparticles is a distinct symptom of damage (Boran *et al.*, 2012). Hyaline degeneration was observed in the liver of carp treated with citrate-capped silver nanoparticles (Lee *et al.*, 2012). In the present study no hyaline degeneration was found in hepatocytes of fish exposed to AgNPs, while distinct symptoms of progressive hyaline degeneration occurred in hepatocytes of fish treated with AgNPs.

Exposure of metal silver to freshwater fish *Cyprinus carpio* in the aquatic environment, hepatocellular necrosis has been observed (Olojo *et al.*, 2005). In liver of *Fundulus heteroclitus* according to Schmalz *et al.*, (2002) the presense of necrotic areas exposed to mercury and other nanoparticles was related to the increase of an inflammatory response and formation of MMCs. In the stressed animals vacuoles in the cytoplasm of the hepatocytes contain lipids and glycogen (Hinton and Lauren, 1991), because the glycogen acts as a reserve of glucose to supply the higher energetic demand occurring in such situations (Panepucci *et al.*, 2001). Vacuolization of the hepatocytes as a single of degenerative water, presence of mainly glycogen in the vacuoles of the hepatocytes, however suggests that vacuolization need not be related to be a degenerative process. The vacuolization of hepatocytes might indicate an imbalance between the rate of synthesis of substances in the parenchymal cells and the rate of their release into the circulation system. In the present study exposure of silver nanoparticles to freshwater fish *Cyprinus carpio* results demonstrate severe damage of hepatic cells and formation of inter cellular spaces could be possibly due to sudden withdrawal and utilization of stored glycogen from the liver cells of fish to meet the energy demand during the exposure of nanoparticles and finally stress occurs.

Histopathological lesions in liver are not specific to pollutants; nevertheless there are numerous reports of histo-cytopathological changes in liver of fish exposed to a wide range of organic compounds and heavy metals like silver nanoparticles (Jimenez-Tenorio *et al.*, 2007). Non-neoplastic lesions, such as cellular and nuclear polymorphism have been considered as an initial toxicopathic lesion resulting from exposure to toxic agents (Stentiford *et al.*, 2003). Agents that produce liver injury can be placed in two types of hepatitis one is toxic hepatitis, another one drug induced hepatitis due to agents that cause liver injury. According to Plaa (1980), the substances that produce toxic hepatitis have the following common characteristics: (1) they produce a distinctive lesion, (2) the severity of the lesions seems to be related to the dose administered (3) while quantitative differences in potency can be found among individuals, generally the same lesion can be produced in all individuals and (4) they usually appear after a predictable, usually brief, latent period. The factors which are

involved in liver injury are biotransformation of toxicants, alteration of hepatic blood flow and potentiating of hepatotoxicity. The occurrence percentages of the histopathological changes in the livers are more widespread than those in the kidneys (Ayas *et al.*, 2007).

The contaminated water affects the kidney of teleost (Thophon *et al.*, 2003). In the present study control fish showed normal structure of renal tubules and haemopoietic tissue. Exposure to nanoparticles frequently causes alterations in the tubules and glomerulus in perch *Lates calcarifer* exposed to cadmium (Thophon *et al.*, 2003). Handy and Penrice (1993) noticed swollen Bowman's capsule cells and melanomacrophages in the kidney of trout *Salmo trutta* and *Oreochromis mossambicus* affected by mercuric chloride and nanoparticles. Fish mullet *Valamugil georgii* exposed to nitrite, showed tubule regeneration and dilation of capillaries in glomerulus. Similarly, *Oncorhynchus mykiss* exposed to nitrite causes alterations in glomerulus and tubules (Bucher and Hofer 1993). Pathological symptoms like dilation of renal tubules and reduction of lymphoid tissue in the kidney of *Lebistes reticulatus* in response to prolonged (3 months) exposure to lead. Renal epithelial necrosis, sloughing of the epithelium, and accumulation of necrotic debris within lumen of the renal tubules in *Mugil auratus* exposed to organic and inorganic mercury. According to severe pathological changes noticed in the kidney may be due to renal excretion of toxicant silver nanoparticles (Tilak *et al.*, 2001).

Renal excretion is one of the ways of eliminating the under-toxified toxicant molecule resulting in severe pathological changes in haemopoietic tissues resulting severe necrosis, cloudy smearing of renal tubules, disintegration of interstitial tissue, pyknotic nuclei (Tilak *et al.*, 2001). A toxicologic insult to the kidney could affect any or all these functions; the cellular response to a toxic insult may vary from an imperceptible biochemical aberration to cell death with resulting necrosis; functionally, toxicity may be reflected as a minor alteration in transport capability or as renal failure (Hook 1980). Depending on the magnitude of the insult, these changes may be reversible, permanent or lethal. Kidney is under the influence of the sympathetic nervous system and changes in neural activity can markedly influence renal function; renal nerve activity might directly influence proximal tubular functions as well; any change in systemic homeostasis that would alter sympathetic nerve activity could also influence the kidney (Hook 1980).

The effects are far more pronounced in acute condition than in chronic exposure. It can be stated that in chronic conditions of exposure some sort of repair mechanism takes

places in fish kidney (Ahmad and Srivastava 1995). A single dose of nephrotoxin agent may produce profound changes in renal function; similarly, chronic administration of a low dose of a nephrotoxin chemical may bring about significant changes in structure of the kidney. Even sublethal concentrations of a toxicant, which lack the capacity to dramatically increase the rate of mortality in exposed populations, may still cause their ecological death after a sufficiently long time of exposure, probably as a result of the cumulative impact of impaired metabolic functions which may explain the enhanced nephral damage of fish from sublethal toxicity in the present study. In general fish seems to acclimatize with sublethal silver and silver nanoparticles exposure in chronic conditions. In the present study shows, in the kidney of fish *Cyprinus carpio*, glomerular expansions, swelling of tubules, dilation of capillaries in the clomerulus, hyaline droplet degeneration are produced by acute and suplethal treatment of silver nanoparticles. The toxic action of nanoparticles may be direct on the cellular components and induce stress responses.

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## SUMMARY

The median lethal concentration (LC50) of silver nanoparticles to an Indian major carp, *Cyprinus carpio* for 24 and 96h were found to be 1.0 and 0.5 mg/L respectively. 1/10<sup>th</sup> of the LC50 value for 24h (0.01mg/L) was taken as sublethal concentration. Epithelial tissue in fish gills forms the half of the body surface area. The current study illustrates that gills are affected due to accumulation of silver nanoparticles which in turn causes epithelial lifting, hyperplasia and hypertrophy of epithelial cells. Nano particle exposure creates changes in the fish hormone level and hence arises metabolic problems. The heavy metals accumulate in kidney changes were noticed and may be due to renal excretion of toxicant molecule resulting in severe pathological changes in hemopoietic tissues resulting severe necrosis.

*Cyprinus carpio* reported a decrease in the blood glucose level when exposed to sublethal concentration of silver nanoparticles which might be due to induction of hypoglycemia in the fish or due to the stress of toxicant. Significant decrease of glycogen may be due to the absence of metabolite named plasmatic pyruvate in liver. Current study reveals histological damage to gill surfaces by silver nanoparticles is attributed to high accumulations in gills, irritation due to elevated mucous secretion, increased ventilation volume and decreased gill oxygen uptake efficiency. Gill hyperplasia might serve as a defensive mechanism leading to a decrease in the respiratory surface and an increase in the toxicant blood diffusion distance.

Biotransformation of toxicant and alteration of hepatic blood flow and potentiation of hepatotoxicity causes hepatic cell damage in liver or it might be due to distinctive lesions and the severity of the lesions seems to be related to the dose administered potency of silver nanoparticles. In the present study during acute and sublethal treatment of silver nanoparticles the kidney of fish *Cyprinus carpio* shows glomerular expansions, swelling of tubules, dilation of capillaries in the glomerular and hyaline droplet degeneration or it might be due to toxic action of silver nanoparticles on the cellular components and induces stress response in cells.

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